

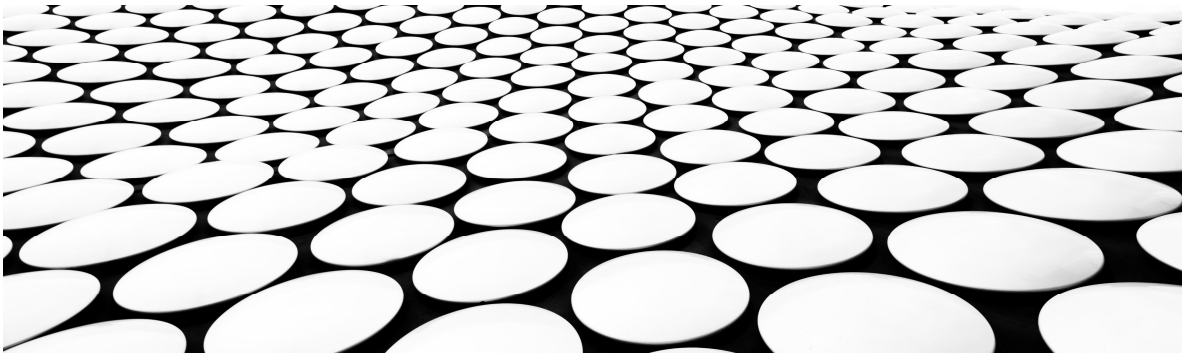


ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH  
**TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN**  
KHOA ĐIỆN TỬ - VIỄN THÔNG

# THIẾT KẾ VI MẠCH ĐIỆN TỬ

## CHAPTER 4: ANALOG INTEGRATED CIRCUIT DESIGN

LÊ ĐỨC HÙNG



### CONTENT

- CMOS Analog circuits design

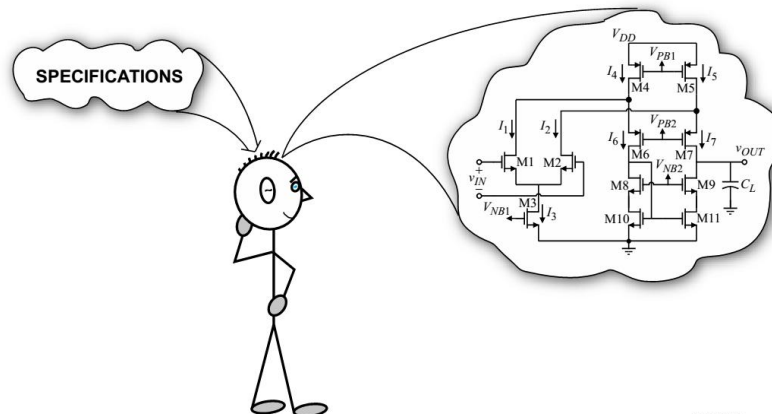
## THI GIỮA KỲ

- Tuần 8: 29/6/2022 (T4)
- Thời gian: 60'
- Nội dung:
  - Quy trình chế tạo CMOS
  - Cấu trúc MOSFET
  - Tính điện thế, dòng điện mối nối PN và MOSFET
- Không nghỉ, học 2 tiếng, rồi thi giữa kỳ

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## INTRODUCTION

- This course teaches Integrated Circuit (IC) design using CMOS technology.
- The objective is to **know W and L of each transistor**.

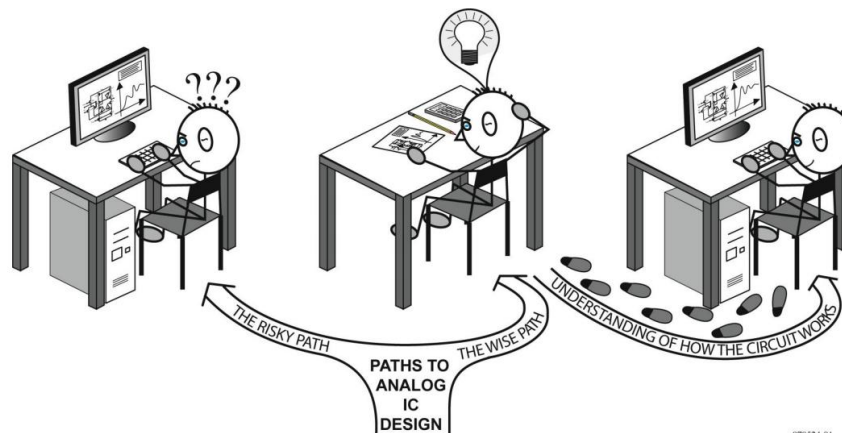


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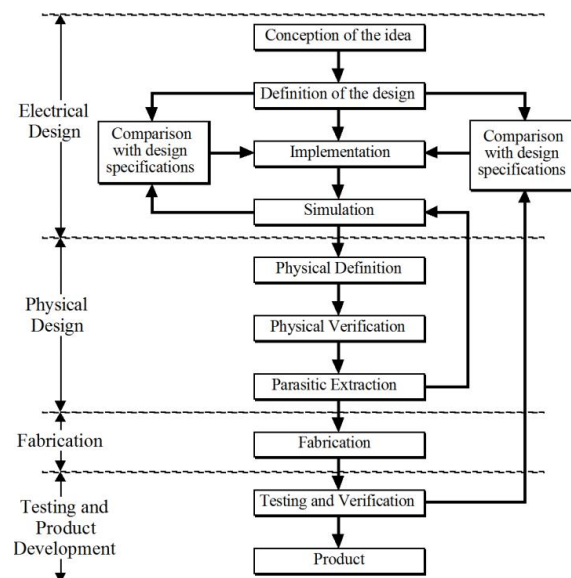
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## INTRODUCTION

- Although simulators are very powerful, the designer must understand the circuit before using the computer to simulate a circuit.



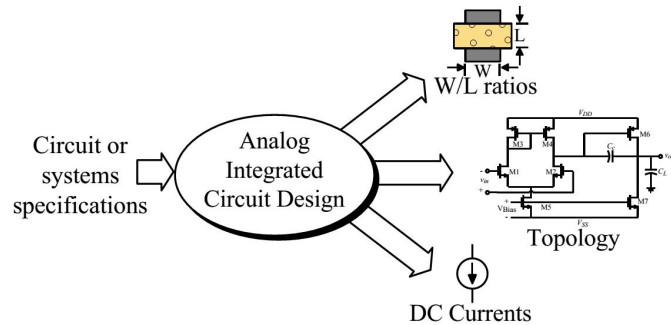
## ANALOG IC DESIGN PROCESS



## INTRODUCTION

### ■ What is Electrical Design ?

- Electrical design is the process of going from the specifications to a circuit solution. The inputs and outputs of electrical design are:



- The electrical design requires active and passive device electrical models for
  - Creating the design
  - Verifying the design
  - Determining the robustness of the design

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## INTRODUCTION

### Steps in Electrical Design

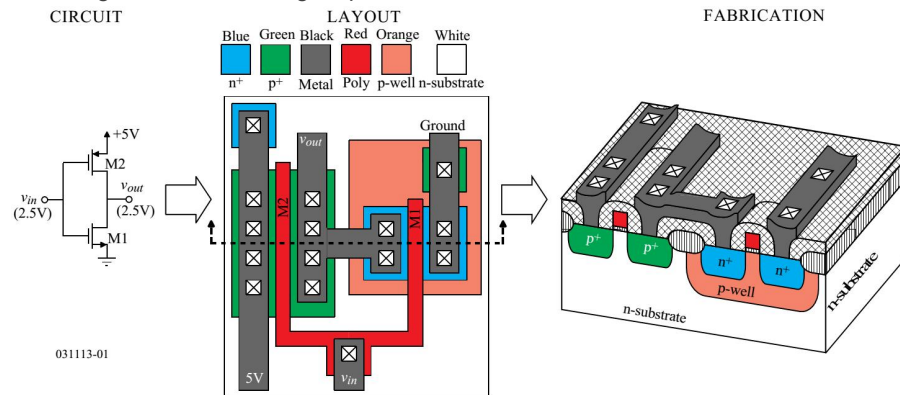
1. Selection of a solution
  - Examine previous designs
  - Select a solution that is simple
2. Investigate the solution
  - Analyze the performance (without a computer)
  - Determine the strengths and weaknesses of the solution
3. Modification of the solution
  - Use the key principles, concepts and techniques to implement
  - Evaluate the modifications through analysis (still no computers)
4. Verification of the solution
  - Use a simulator with precise models and verify the solution
  - Large disagreements with the hand analysis and computer verification should be carefully examined.

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## INTRODUCTION

### What is Physical Design ?

- Physical design is the process of representing the electrical design in a layout consisting of many distinct geometrical rectangles at various levels. The layout is then used to create the actual, three-dimensional integrated circuit through a process called fabrication.



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## INTRODUCTION

### What is the Layout Process?

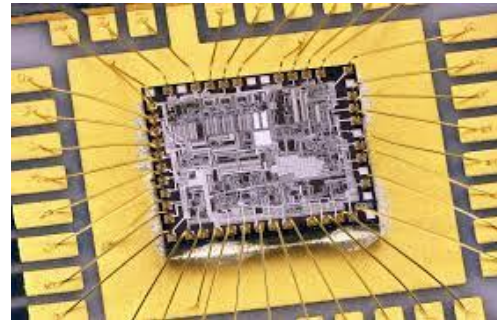
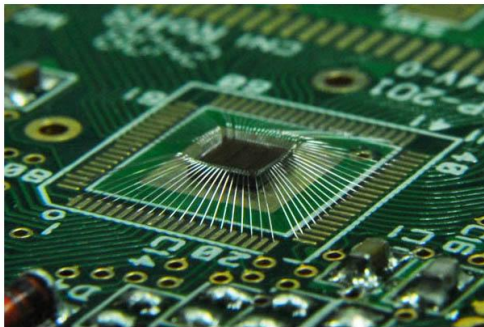
- Inputs are the W/L values and the schematic (generally from schematic entry used for simulation).
- A CAD tool is used to enter the various geometries. The designer must enter the location, shape, and level of the particular geometry.
- During the layout, the designer must obey a set of rules called design rules (DRC). These rules are for the purpose of ensuring the robustness and reliability of the technology.
- Once the layout is complete, then a process called layout versus schematic (LVS) is applied to determine if the physical layout represents the electrical schematic.
- The next step is now that the physical dimensions of the design are known, the parasitics can be extracted. These parasitics primarily include:
  - Capacitance from a conductor to ground
  - Capacitance between conductors
  - Bulk resistance
- The extracted parasitics are entered into the simulated database and the design is resimulated to insure that the parasitics will not cause the design to fail.

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## INTRODUCTION

### Packaging

Wire bonding



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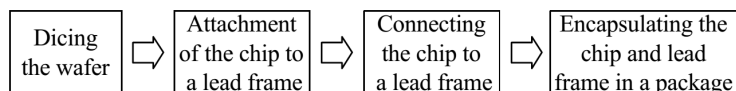
## INTRODUCTION

### Packaging

- Packaging of the integrated circuit is an important part of the physical design process. The function of packaging is:

1. Protect the integrated circuit
2. Power the integrated circuit
3. Cool the integrated circuit
4. Provide the electrical and mechanical connection between the integrated circuit and the outside world.

- Packaging steps:



- Other considerations of packaging:

1. • Speed
2. • Parasitics (capacitive and inductive)

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## INTRODUCTION

### What is Test Design ?

- Test design is the process of coordinating, planning and implementing the measurement of the analog integrated circuit performance.
- Objective: To compare the experimental performance with the specifications and/or simulation results.
- Types of tests:
  - Functional – verification of the nominal specifications
  - Parametric – verification of the characteristics to within a specified tolerance
  - Static – verification of the static (AC and DC) characteristics of a circuit or system
  - Dynamic – verification of the dynamic (transient) characteristics of a circuit or system
- Additional Considerations:
  - Should the testing be done at the wafer level or package level?
  - How do you remove the influence (de-embed) of the measurement system from the measurement?

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## INTRODUCTION

### Characteristics of Analog Integrated Circuit Design

- Done at the circuits level
- Complexity is high
- Continues to provide challenges as technology evolves
- Demands a strong understanding of the principles, concepts and techniques
- Good designers generally have a good physics background
- Must be able to make appropriate simplifications and assumptions
- Requires a good grasp of both modeling and technology
- Have a wide range of skills - breadth (analog only is rare)
- Be able to learn from failure
- Be able to use simulation correctly

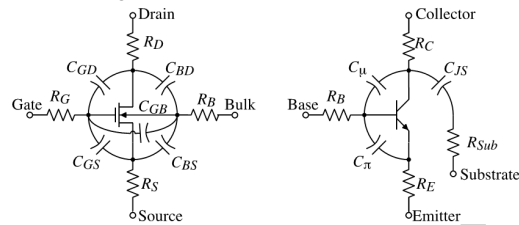
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## INTRODUCTION

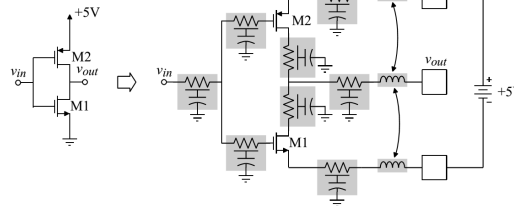
### Understanding Technology

- Understanding technology helps the analog IC designer to know the limits of the technology and the influence of the technology on the design.

- Device Parasitics:



- Connection Parasitics:

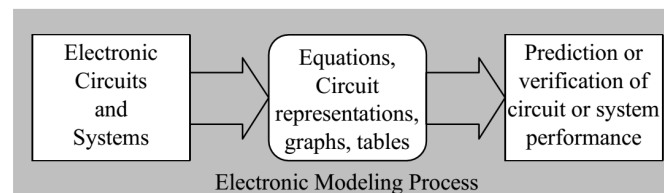


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## INTRODUCTION

### Understanding Modeling

- Modeling is the process by which the electrical properties of an electronic circuit or system are represented by means of mathematical equations, circuit representations, graphs or tables.
- Models permit the predicting or verification of the performance of an electronic circuit or system.



- Examples: Ohm's law, the large signal model of a MOSFET, the I-V curves of a diode, etc.
- Goal: Models that are simple and allow the designer to understand the circuit performance.

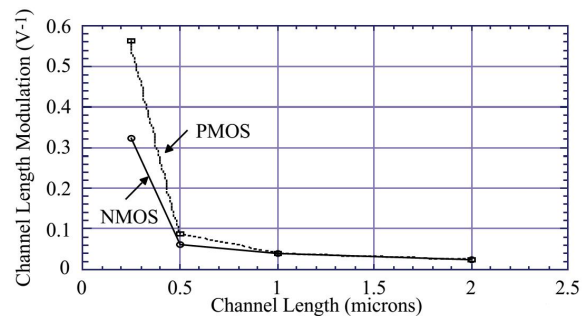
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## INTRODUCTION

### Understanding Modeling: Influence of Channel Length on $\lambda$

- Note that the value of  $\lambda$  varies with channel length,  $L$ . The data below is from a 0.25 $\mu\text{m}$  CMOS technology.



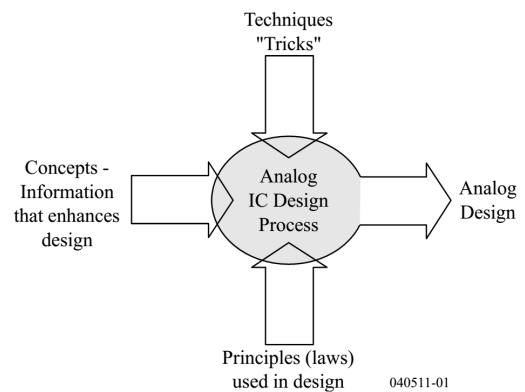
- Most analog designers stay away from minimum channel length to get better gains and matching at the sacrifice of speed.

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## INTRODUCTION

### Key Principles, Concepts and Techniques of Analog IC Design

- Principles mean fundamental laws that are precise and never change.
- Concepts will include relationships, "soft-laws" (ones that are generally true), analytical tools, things worth remembering.
- Techniques will include the assumptions, "tricks", tools, methods that one uses to simplify and understand.



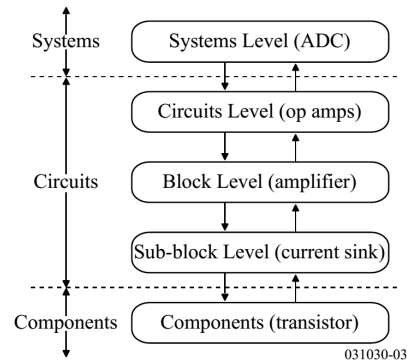
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## INTRODUCTION

### Complexity in Analog Design

- Analog design is normally done in a non-hierarchical manner and makes little use of repeated blocks. As a consequence, analog design can become quite complex and challenging.
- How do you handle the complexity?
  1. Use as much hierarchy as possible.
  2. Use appropriate organization techniques.
  3. Document the design in an efficient manner.
  4. Make use of assumptions and simplifications.
  5. Use simulators appropriately.



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## INTRODUCTION

### Assumptions

- An assumption is taking something to be true without formal proof. Assumptions in analog circuit design are used for simplifying the analysis or design. The goal of an assumption is to separate the essential information from the nonessential information of a problem.
- The elements of an assumption are:
  1. Formulating the assumption to simplify the problem without eliminating the essential information.
  2. Application of the assumption to get a solution or result.
  3. Verification that the assumption was in fact appropriate.
- Examples:
  - Neglecting a large resistance in parallel with a small resistance
  - Miller effect to find a dominant pole
  - Finding the roots of a second-order polynomial assuming the roots are real and separated

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## INTRODUCTION

### Analog IC design today has reached maturity (almost developed)

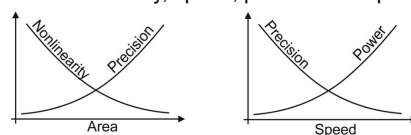
- There are established fields of application:
  - Digital-analog and analog-digital conversion (ADC, DAC)
  - Disk drive controllers
  - Modems - filters
  - Bandgap reference
  - Analog phase lock loops
  - DC-DC conversion
  - Buffers
  - Codecs, Etc.
- Existing philosophy regarding analog circuits: "If it can be done economically by digital, don't use analog."
- Consequently: Analog finds applications where speed, area, or power have advantages over a digital approach.

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## INTRODUCTION

### Analog IC design Challenges

- Technology:
  - Digital circuits have scaled well with technology
  - Analog does not benefit as much from smaller features
    - Speed, gain, matching increases
    - Nonlinearity increases
    - New issues appear such as gate current leakage
- Analog Circuit Challenges:
  - Trade offs are necessary between linearity, speed, precision and power



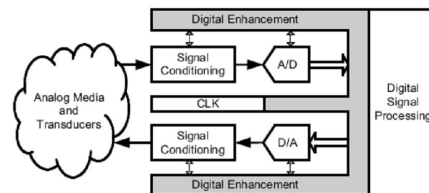
- As analog is combined with more digital, substrate interference will become worse

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## INTRODUCTION

### Digitally assisted Analog Circuits

- Use digital circuits which work better at scaled technologies to improve analog circuits that do not necessarily improve with technology scaling.
- Principles and Techniques:
  - Open-loop vs. closed loop
    - Open loop is less accurate but smaller → Faster, less power
    - Closed-loop is more accurate but larger → Slower, more power
  - Averaging
    - Increase of accuracy → Smaller devices, more speed
  - Calibration
    - Accuracy increases → Increased resolution with same area
  - Dynamic Element Matching
    - Enhancement of component precision
  - Doubly correlated sampling
    - Reduction of dc influences (noise, offset) → Smaller devices, more speed
  - Etc.



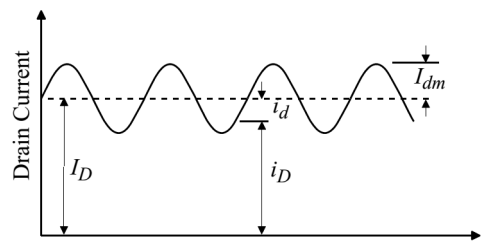
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## NOTATION, TERMINOLOGY AND SYMBOLOGY

### Definition of Symbols for Various Signals

| Signal Definition                                   | Quantity  | Subscript | Example |
|-----------------------------------------------------|-----------|-----------|---------|
| Total instantaneous value of the signal             | Lowercase | Uppercase | $q_A$   |
| DC value of the signal                              | Uppercase | Uppercase | $Q_A$   |
| AC value of the signal                              | Lowercase | Lowercase | $q_a$   |
| Complex variable, phase, or rms value of the signal | Uppercase | Lowercase | $Q_a$   |

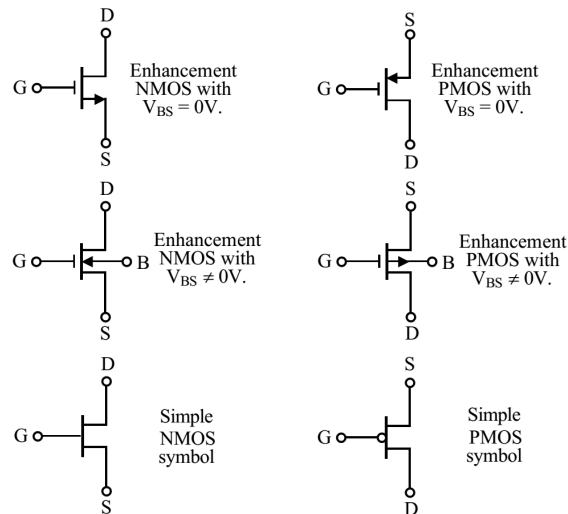
- Example:



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## NOTATION, TERMINOLOGY AND SYMBOLOGY

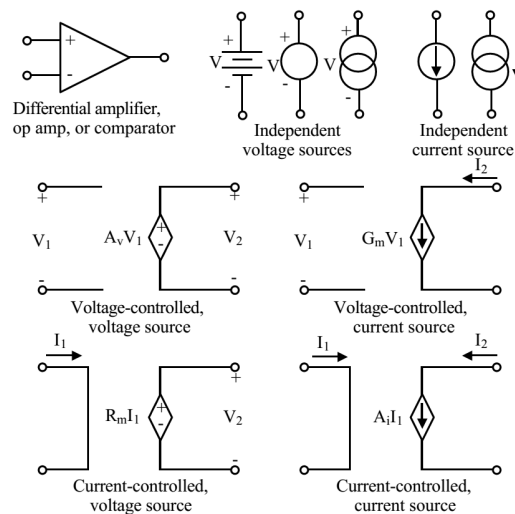
### MOS Transistor Symbols



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## NOTATION, TERMINOLOGY AND SYMBOLOGY

### Other Schematic Symbols



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## MOSFET SWITCH

### Switch Model

- An ideal switch is a short-circuit when ON and an open-circuit when OFF.  $V_C$  = controlling terminal for the switch ( $V_C$  high  $\rightarrow$  switch ON,  $V_C$  low  $\rightarrow$  switch OFF)

- Actual switch:

$r_{on}$  = resistance of the switch when ON

$r_{off}$  = resistance of the switch when OFF

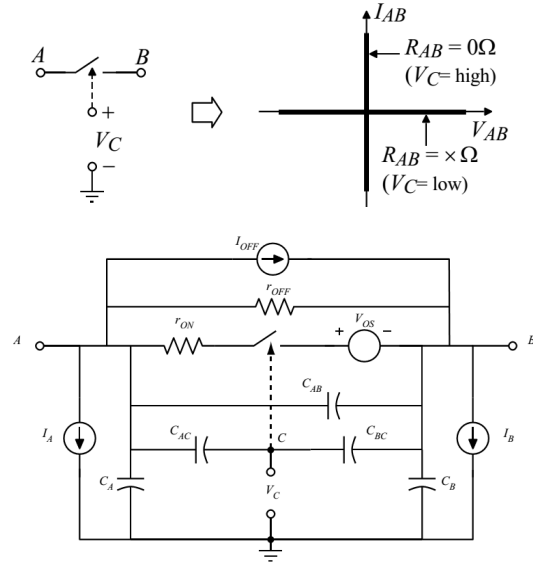
$V_{OS}$  = offset voltage when the switch is ON

$I_{off}$  = offset current when the switch is OFF

$I_A$  and  $I_B$  are leakage currents to ground

$C_A$  and  $C_B$  are capacitances to ground

$C_{AC}$  and  $C_{BC}$  = parasitic capacitors between the control terminal and switch terminals



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## MOSFET SWITCH

### MOS Transistor as a Switch



- ON Characteristics of a MOS Switch

Assume operation in active region ( $v_{DS} < v_{GS} - V_T$ ) and  $v_{DS}$  small.

$$i_D = \frac{\mu C_{ox} W}{L} \left[ (v_{GS} - V_T) - \frac{v_{DS}}{2} \right] v_{DS} \approx \frac{\mu C_{ox} W}{L} (v_{GS} - V_T) v_{DS}$$

Then

$$R_{ON} \approx \frac{v_{DS}}{i_D} = \frac{1}{\frac{\mu C_{ox} W}{L} (v_{GS} - V_T)} \quad \beta = \frac{\mu C_{ox} W}{L}$$

- OFF Characteristics of a MOS Switch

If  $v_{GS} < V_T$ , then  $i_D = I_{OFF} = 0$  when  $v_{DS} \approx 0V$ .

If  $v_{DS} > 0$ , then

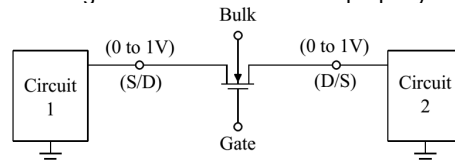
$$R_{OFF} \approx \frac{1}{i_D \lambda} = \frac{1}{I_{OFF} \lambda} \approx \infty$$

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## MOSFET SWITCH

### MOS Switch Voltage Ranges

- If a MOS switch is used to connect two circuits that can have analog signal that vary from 0 to 1V, what must be the value of the bulk and gate voltages for the switch to work properly?



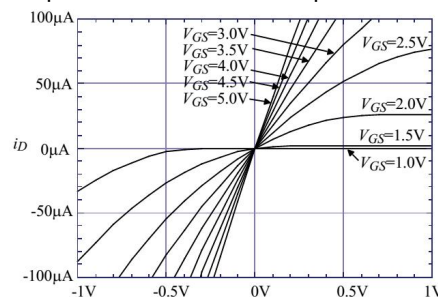
- To insure that the bulk-source and bulk-drain pn junctions are reverse biased, the bulk voltage must be less than the minimum analog signal for a NMOS switch.
- To insure that the switch is ON, the gate voltage must be greater than the maximum analog signal + the threshold for a NMOS switch.
- Therefore:
  - $V_{Bulk} \leq 0V$
  - $V_{Gate(on)} > 1V + V_T$
  - $V_{Gate(off)} \leq 0V$
- However, the large value of reverse bias bulk voltage causes the threshold voltage to increase.

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## MOSFET SWITCH

### Current-Voltage Characteristics of a NMOS Switch

- The following simulated output characteristics correspond to triode operation of the MOSFET.



- SPICE Input File

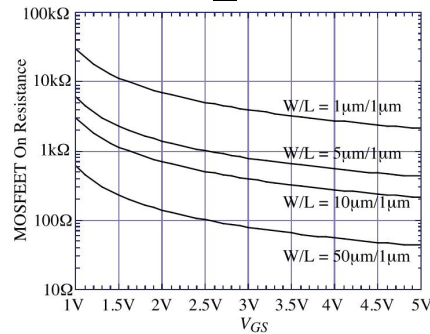
```
MOS Switch On Characteristics
M1 1 2 0 3 MNMOS W=1U L=1U
.MODEL MNMOS NMOS VTO=0.7, KP=110U,
+LAMBDA=0.04, GAMMA=0.4 PHI=0.7
VDS 1 0 DC 0.0
```

```
VGS 2 0 DC 0.0
VBS 3 0 DC -5.0
.DC VDS -1 1 0.1 VGS 1 5 0.5
.PRINT DC ID(M1)
.PROBE
.END
```

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## MOSFET SWITCH

### MOS Switch ON Resistance as a function of $V_{GS}$



#### SPICE Input File

MOS Switch On Resistance as a f(W/L)

```
M1 1 2 0 0 MNMOS W=1U L=1U
M2 1 2 0 0 MNMOS W=5U L=1U
M3 1 2 0 0 MNMOS W=10U L=1U
M4 1 2 0 0 MNMOS W=50U L=1U
.MODEL MNMOS NMOS VTO=0.7, KP=110U,
```

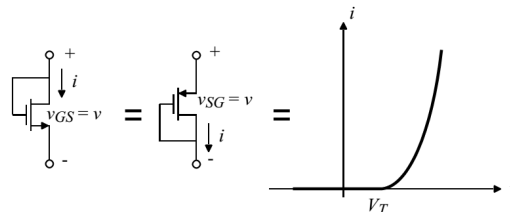
```
+LAMBDA=0.04, GAMMA=0.4, PHI=0.7
VDS 1 0 DC 0.001V
VGS 2 0 DC 0.0
.DC VGS 1 5 0.1
.PRINT DC ID(M1) ID(M2) ID(M3) ID(M4)
.PROBE
.END
```

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## MOSFET DIODE

### MOS Diode

- When the MOSFET has the gate connected to the drain, it acts like a diode with characteristics similar to a pn-junction diode.



- When the gate is connected to the drain of an enhancement MOSFET, the MOSFET is always in the saturation region.

$$v_{DS} \geq v_{GS} - V_T \rightarrow v_D - v_S \geq v_G - v_S - V_T \rightarrow v_D - v_G \geq -V_T \rightarrow v_{DG} \geq -V_T$$

- Since  $V_T$  is always greater than zero for an enhancement device, then  $v_{DG} = 0$  satisfies the conditions for saturation.

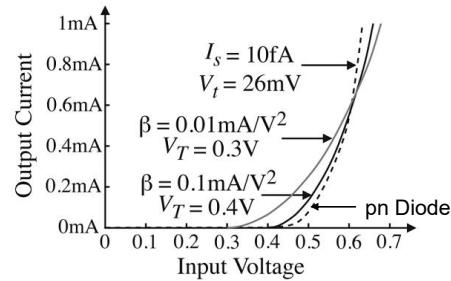
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## MOSFET DIODE

### How Does the MOS Diode compare with a PN Diode?

- The comparison is basically the difference between an exponential and a square-law function. However, if the designer is willing to spend  $W/L$ , the comparison becomes more interesting as shown below.



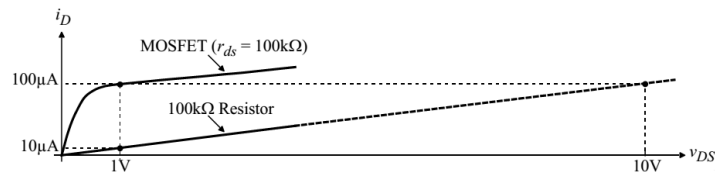
- If the threshold voltage is less than 0.4V, the MOS diode can provide more current for the same voltage than a PN junction diode even for modest  $W/L$  ratios. However, at the same value of current, the PN junction always has a larger transconductance.

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## MOSFET RESISTORS

### Real Resistors vs. MOSFET Resistors

- A MOS resistor is smaller in area than a real resistor, but its linearity is the primary problem.
- Can pass a large current through a large resistance without a large voltage drop



- AC resistance of MOS. Resistors  $= \frac{v_{ds}}{i_d} = \frac{1}{g_{ds}}$   
 where  $g_{ds} \approx \frac{\beta}{2} (V_{GS} - V_T)^2 \lambda = I_D \lambda$

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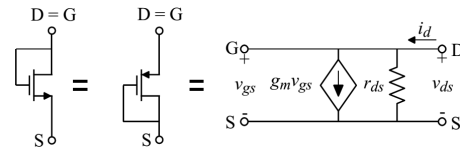
## MOSFET RESISTORS

### MOS Diode as a Resistor

- DC resistance

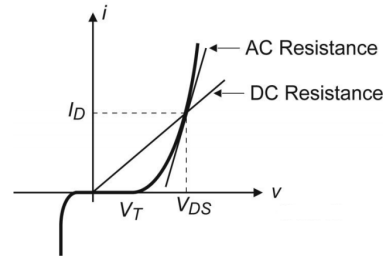
$$\text{DC resistance} = \frac{v_{DS}}{I_D} = \frac{V_T}{I_D} + \sqrt{\frac{2}{\beta I_D}}$$

- AC resistance (Small-Signal Load)



$$\text{AC resistance} = \frac{v_{ds}}{i_d} = \frac{1}{g_m + g_{ds}} \approx \frac{1}{g_m}$$

$$\text{where } g_{ds} \approx \beta(V_{GS} - V_T) = \sqrt{2\beta I_D}$$

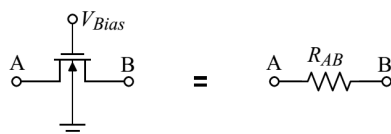


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## MOSFET RESISTORS

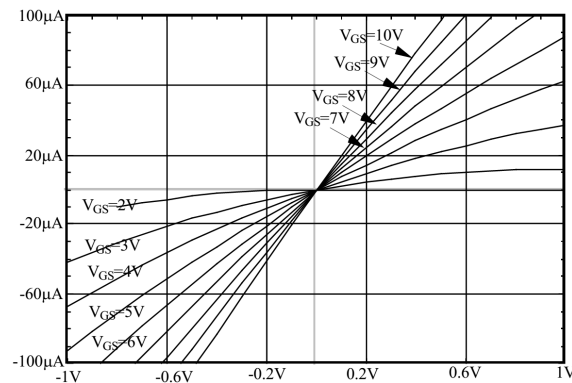
### Use of the MOSFET to Implement a Floating Resistor

- In many applications, it is useful to implement a resistance using a MOSFET. First, consider the simple, single MOSFET implementation.



$$R_{AB} = \frac{L}{K'W(V_{GS} - V_T)}$$

[Review  $R_{ON}$  (Slide 28)]

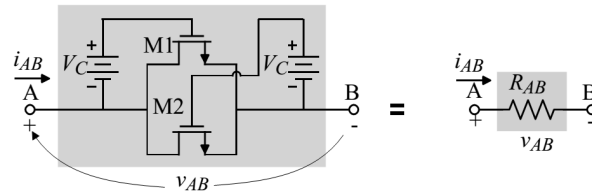


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## MOSFET RESISTORS

### Parallel MOSFET

- Circuit:



- Assume both devices are non-saturated

$$i_{D1} = \beta_1 \left[ (v_{AB} + V_C - V_T)v_{AB} - \frac{v_{AB}^2}{2} \right]$$

$$i_{D2} = \beta_2 \left[ (V_C - V_T)v_{AB} - \frac{v_{AB}^2}{2} \right]$$

$$i_{AB} = i_{D1} + i_{D2} = \beta \left[ v_{AB}^2 + (V_C - V_T)v_{AB} - \frac{v_{AB}^2}{2} + (V_C - V_T)v_{AB} - \frac{v_{AB}^2}{2} \right]$$

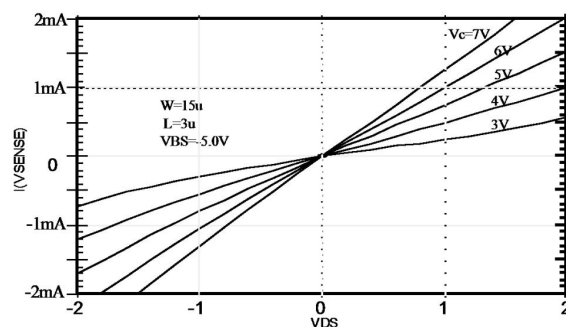
$$i_{AB} = 2\beta(V_C - V_T)v_{AB} \quad \Rightarrow \quad R_{AB} = \frac{1}{2\beta(V_C - V_T)}$$

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## MOSFET RESISTORS

### Parallel MOSFET

- Voltage-Current Characteristic



- SPICE Input File:

```

NMOS parallel transistor realization
M1 2 1 0 5 MNMOS W=15U L=3U
M2 2 4 0 5 MNMOS W=15U L=3U
.MODEL MNMOS NMOS VTO=0.75, KP=25U, +LAMBDA=0.01,
GAMMA=0.8 PHI=0.6
VC 1 2
E1 4 0 1 2 1.0
VSENSE 10 2 DC 0
    
```

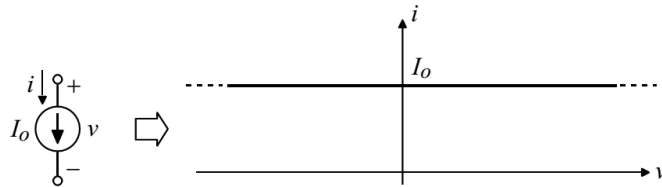
```

VDS 10 0
VSS 5 0 DC -5
.DC VDS -2.0 2.0 .2 VC 3 7 1
.PRINT DC I(VSENSE)
.PROBE
.END
    
```

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## CURRENT SOURCE

### Ideal Current Sources



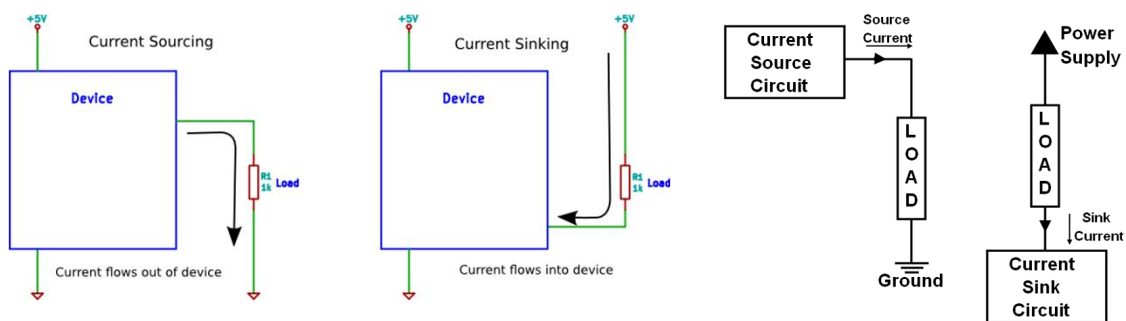
- Current is fixed at a value of  $I_o$
- Voltage can be any values from  $-\infty$  to  $+\infty$
- Be careful when using a current sink or source to replace a MOSFET sink/source in simulation

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## CURRENT SOURCE

### Current Source vs Sink

- Source current refers to the ability of the digital output port to *supply* current. Sink current refers to the ability of the port to *receive* current.



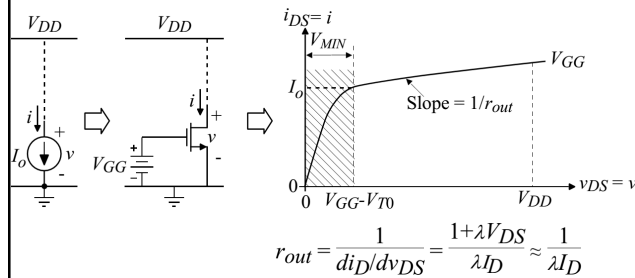
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## CURRENT SOURCE

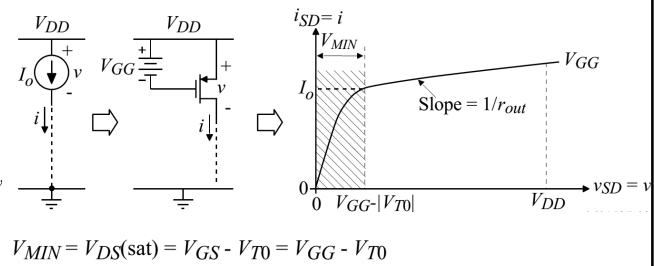
### Characterization of MOSFET Sinks and Sources

A sink/source is characterized by two quantities:

- $r_{out}$  - a measure of the "flatness" of the current sink/source (its independence of voltage)
- $V_{MIN}$  - the min. across the sink or source for which the current is no longer constant
- NMOS Current Sink:



### PMOS Current Sink:



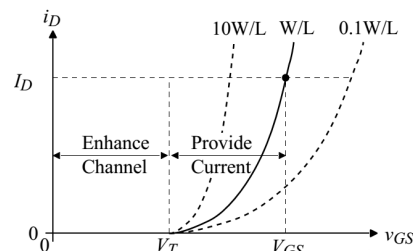
- Note: The NMOS current sink can only have positive values of  $v$ .

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## CURRENT SOURCE

### Gate-Source Voltage Components

- It is important to note that the gate-source voltage consists of two parts as illustrated below:



$$V_{GS} = V_{T0} + V_{ON} = \text{Part to enhance the channel} + \text{Part to cause current flow}$$

where

$$V_{ON} = V_{DS(sat)} = V_{GS} - V_{T0}$$

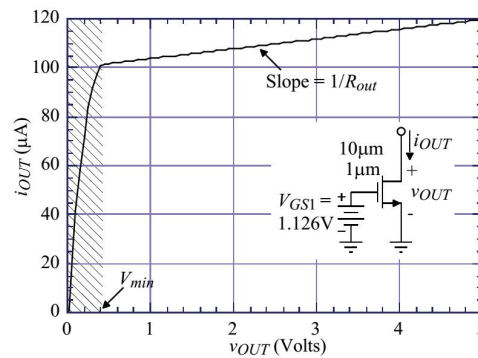
$$\therefore V_{MIN} = V_{ON} = V_{DS(sat)} = \sqrt{\frac{2I_D}{K'(W/L)}} \quad (\text{For the simple current sink})$$

- Note that  $V_{MIN}$  can be reduced by using large values of  $W/L$ .

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## CURRENT SOURCE

### Simulation of a Simple MOS Current Sink



#### Comments:

- $V_{MIN}$  is too large  $\rightarrow$  desire  $V_{MIN}$  to approach zero
- Slope too high  $\rightarrow$  desire the characteristic to be flat implying very large output resistance

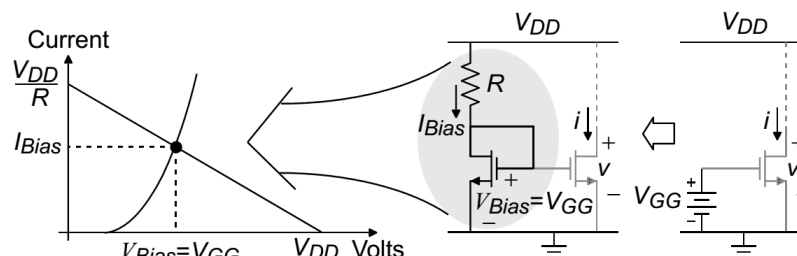
$$(K_N' = 110\mu A/V^2, V_T = 0.7V \text{ and } \lambda = 0.04V^{-1}) \rightarrow r_{ds} = 250k\Omega$$

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## CURRENT SOURCE

### How is VGG Implemented?

- The only voltage source assumed available is  $V_{DD}$ .
- Therefore,  $V_{GG} = V_{bias}$  can be implemented in many ways with the example below being one way.



- Better and more stable implementations of VGG will be shown later.

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## CURRENT SOURCE

### Improving the Performance of the Simple NMOS Current Sink

- The simple NMOS current sink shown previously (Slide 41) had two problems.
  1. The value of  $V_{MIN}$  may be too large.
  2. The output resistance ( $250k\Omega$ ) was too small.
- How can the designer solve these problems?
  1. The first problem can be solved by **increasing the W/L value** of the NMOS transistor.

$$V_{MIN} = V_{ON} = V_{DS(sat)} = \sqrt{\frac{2I_D}{K'(W/L)}}$$

- In the simulation shown previously,

$$V_{MIN} = \sqrt{\frac{2 \cdot 100\mu A}{110\mu A/V^2 \cdot 10}} = 0.426V$$

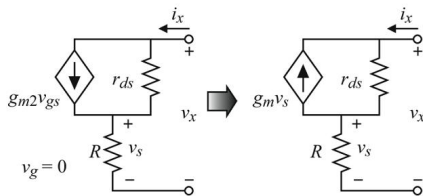
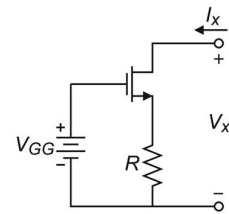
- We could decrease this to 0.1V with a  $W/L = 182$ .
- 2. How can **the small output resistance be increased**? → Answer is **feedback**.

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## CURRENT SOURCE

### Increasing the Output Resistance of the Simple Current Sink

- Choosing series feedback, we select the following circuit to boost the output resistance of the simple current sink:
- Assume that we can neglect the bulk effect and find the output resistance by small-signal analysis



$$v_x = (i_x + g_m v_s) r_{ds} + i_x R$$

$$v_x = (i_x + g_m i_x R) r_{ds} + i_x R = i_x (r_{ds} + R + g_m r_{ds} R)$$

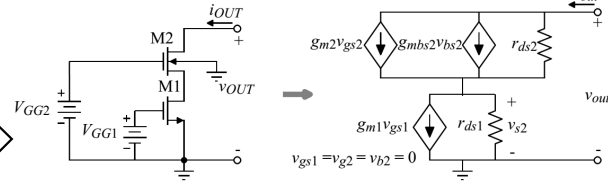
$$\therefore R_x = \frac{v_x}{i_x} = r_{ds} + R + g_m r_{ds} R \approx g_m r_{ds} R$$

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## CURRENT SOURCE

### Cascode Current Sink

- Replacing R with the simple current sink leads to a practical implementation shown as:



### Small-signal output resistance:

- Noting that  $v_{gs1} = v_{gs2} = v_{bs2} = 0$  and writing a loop equation we get,

$$v_{out} = (i_{out} - g_{m2}v_{gs2} - g_{mbs2}v_{bs2})r_{ds2} + r_{ds1}i_{out}$$

- However,

$$v_{gs2} = 0 - v_{s2} = -i_{out}r_{ds1} \text{ and } v_{bs2} = 0 - v_{s2} = -i_{out}r_{ds1}$$

Therefore,

$$v_{out} = i_{out}[r_{ds1} + r_{ds2} + g_{m2}r_{ds1}r_{ds2} + g_{mbs}r_{ds1}r_{ds2}]$$

or

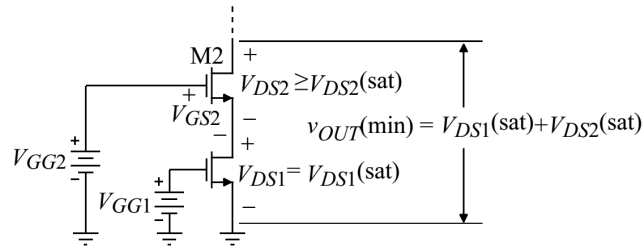
$$r_{out} = \frac{v_{out}}{i_{out}} = r_{ds1} + r_{ds2} + g_{m2}r_{ds1}r_{ds2} + g_{mbs}r_{ds1}r_{ds2} \approx g_{m2}r_{ds1}r_{ds2}$$

- A general principle is beginning to emerge: The output resistance of a cascode circuit  $\approx R \times$  (Common source voltage gain of the cascoding transistor)

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## CURRENT SOURCE

### Design of $V_{GG1}$ and $V_{GG2}$



- $V_{GG1}$  is selected to provide the desired current. M1 is assumed to be in saturation.
- $V_{GG2}$  is selected to keep  $V_{DS1}$  as small as possible and still be in saturation.

$$V_{GG2} = V_{DS1}(\text{sat}) + V_{GS2} = V_{DS1}(\text{sat}) + V_T + V_{DS2}(\text{sat})$$

$$\text{If } W_1/L_1 = W_2/L_2, \text{ then } V_{GG2} = 2V_{DS}(\text{sat}) + V_T = 2V_{ON} + V_T$$

- Thus, for the previous NMOS current sink,  $V_{GG2}$  would be equal to:

$$V_{GG2} = 2(0.426) + 0.7 = 1.552\text{V}$$

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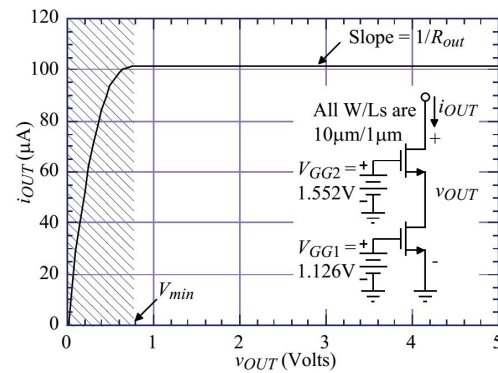
## CURRENT SOURCE

### Example of the cascode CMOS Current Sink

- Example: Use the model parameters  $K_N' = 110 \mu\text{A}/\text{V}^2$ ,  $V_T = 0.7\text{V}$  and  $\lambda_N = 0.04\text{V}^{-1}$  to calculate
  - $r_{ds1}$  and  $r_{ds2}$  for the simple current sink if  $I_{OUT} = 100 \mu\text{A}$
  - the small-signal output resistance for the cascode current sink with  $I_{OUT} = 100 \mu\text{A}$ . Assume that all W/L values are 1.

■ Solution:

- Using  $\lambda = 0.04 \text{ V}^{-1}$  and  $I_{OUT} = 100 \mu\text{A}$  gives  $r_{ds1} = 250\text{k}\Omega = r_{ds2}$ .
- Ignoring the bulk effect, we find that  $g_{m1} = g_{m2} = 469 \mu\text{S}$  which gives  $r_{out} = (250\text{k}\Omega)(469 \mu\text{S})(250\text{k}\Omega) = 29.32\text{M}\Omega$ .



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## CURRENT SOURCE

### Summary of Both BJT and MOS Current Sinks/Sources

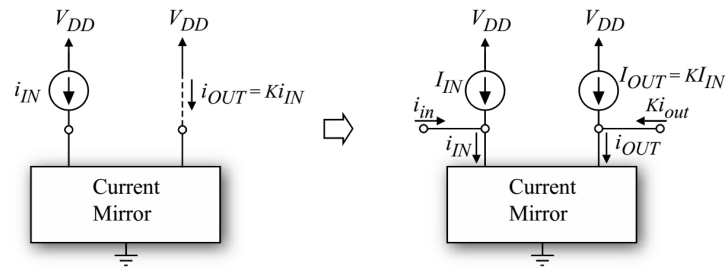
| Current Sink/Source                              | $r_{OUT}$                                                    | $V_{MIN}$                         |
|--------------------------------------------------|--------------------------------------------------------------|-----------------------------------|
| Simple MOS Current Sink                          | $r_{ds} = \frac{1}{\lambda I_D}$                             | $V_{DS(sat)} = V_{ON}$            |
| Simple BJT Current Sink                          | $r_o = \frac{V_A}{I_C}$                                      | $V_{CE(sat)} \approx 0.2\text{V}$ |
| Cascode MOS                                      | $\approx g_{m2} r_{ds2} r_{ds1}$                             | $2V_{ON}$                         |
| Cascode BJT                                      | $\approx \beta_F r_o$                                        | $2V_{CE(sat)}$                    |
| Regulated Cascode Current Sink                   | $\approx r_{ds3} g_{m3} r_{ds2} g_{m4} (r_{ds4}    r_{ds5})$ | $\approx V_T + V_{ON}$            |
| Minimum $V_{MIN}$ Regulated Cascode Current Sink | $\approx r_{ds3} g_{m3} r_{ds2} g_{m4} (r_{ds4}    r_{ds5})$ | $\approx V_{ON}$                  |

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## MOSFET CURRENT MIRRORS

### What is a Current Mirror?

- A current mirror replicates the input current of a current sink or current source as an output current. The output current may be identical to the input current or can be a scaled version of it.



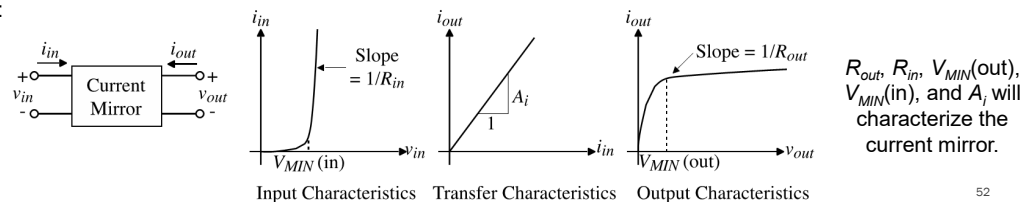
- The above current mirrors are referenced with respect to ground. Current mirrors can also be referenced with respect to  $V_{DD}$  and can source input and output currents.

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## MOSFET CURRENT MIRRORS

### Characterization of Current Mirrors

- A current mirror is basically a **current amplifier**. The ideal characteristics of a current amplifier are:
  - Output current linearly related to the input current,  $i_{out} = A_i i_{in}$
  - Input resistance is zero
  - Output resistance is infinity
- Also, the characteristic  $V_{MIN}$  applies not only to the output but also the input.
  - $V_{MIN}(in)$  is the range of  $v_{in}$  over which the input resistance is not small
  - $V_{MIN}(out)$  is the range of  $v_{out}$  over which the output resistance is not large
- Graphically:

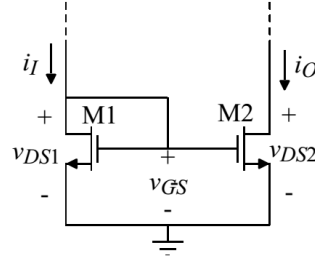


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## MOSFET CURRENT MIRRORS

### A Simple MOS Current Mirror

- Circuit:



- Assume that  $v_{DS2} > v_{GS} - V_{T2}$ , then

$$\frac{i_O}{i_I} = \left( \frac{L_1 W_2}{W_1 L_2} \right) \left( \frac{V_{GS} - V_{T2}}{V_{GS} - V_{T1}} \right)^2 \left[ \frac{1 + \lambda v_{DS2}}{1 + \lambda v_{DS1}} \left( \frac{K_2'}{K_1'} \right) \right]$$

- If the transistors are matched, then  $K_1' = K_2'$  and  $V_{T1} = V_{T2}$  to give,

$$\frac{i_O}{i_I} = \left( \frac{L_1 W_2}{W_1 L_2} \right) \left( \frac{1 + \lambda v_{DS2}}{1 + \lambda v_{DS1}} \right)$$

- If  $v_{DS1} = v_{DS2}$ , then

$$\frac{i_O}{i_I} = \left( \frac{L_1 W_2}{W_1 L_2} \right)$$

- Therefore, the sources of error are:

1.  $v_{DS1} \neq v_{DS2}$
2. M1 and M2 are not matched.

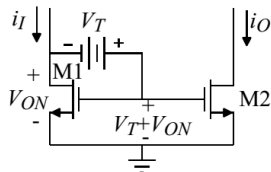


The current mirror circuit needs to be improved

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## MOSFET CURRENT MIRRORS

### Summary of the Simple MOS Current Mirror/Amplifier



- Minimum input voltage is  $V_{MIN}(in) = V_T + V_{ON}$

- Minimum output voltage is  $V_{MIN}(out) = V_{ON}$

- Output resistance is  $R_{out} = \frac{1}{\lambda I_D} = r_{ds}$

- Input resistance is  $R_{in} \approx \frac{1}{g_m}$

- Current gain accuracy is poor because  $v_{DS1} \neq v_{DS2}$

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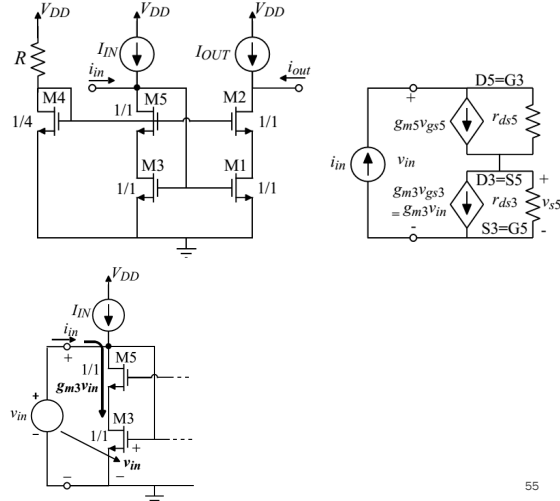
## MOSFET CURRENT MIRRORS

### Improved Current Mirrors - Large Output Swing Cascode Current Mirror

$$R_{out} \approx g_{m2}r_{ds2}r_{ds1}$$

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{r_{ds5} + r_{ds3} + r_{ds3}g_{m5}r_{ds5}}{g_{m3}r_{ds3}(1 + g_{m5}r_{ds5})} \approx \frac{1}{g_{m3}}$$

- An easier way to find  $R_{in}$ :
  1. Apply a small voltage change,  $v_{in}$ , at the input.
  2. Note that this voltage is equal to  $v_{gs3}$ .
  3. This small voltage change causes a current change in the drain of M3 of  $g_{m3}v_{gs3}$  or  $g_{m3}v_{in}$ .
  4. The current  $i_{in}$  is equal to  $g_{m3}v_{in}$ .
  5. Therefore, dividing  $v_{in}$  by  $i_{in}$  gives  $R_{in} = 1/g_{m3}$ .
- $V_{MIN}(out) = 2V_{ON}$
- $V_{MIN}(in) = V_T + V_{ON}$
- Current gain is excellent because  $v_{DS1} = v_{DS3}$



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## MOSFET CURRENT MIRRORS

### Improved Current Mirrors – Self-Biased Cascode Current Mirror

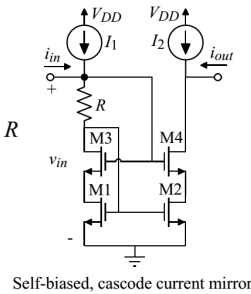
$$v_{in} = i_{in}R + r_{ds3}(i_{in} - g_{m3}v_{gs3}) + r_{ds1}(i_{in} - g_{m1}v_{gs1})$$

$$v_{gs1} = v_{in} - i_{in}R$$

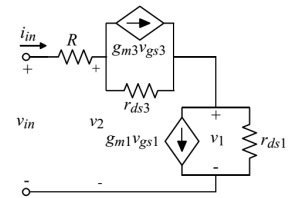
- Input resistance:

$$R_{in} = \frac{R + r_{ds1} + r_{ds3} + g_{m3}r_{ds3}r_{ds1} + g_{m1}r_{ds1}g_{m3}r_{ds3}R}{1 + g_{m3}r_{ds3} + g_{m1}r_{ds1}g_{m3}r_{ds3} + g_{m1}r_{ds1}} \approx \frac{1}{g_{m1}} + R$$

- Output resistance:  $R_{out} \approx g_{m4}r_{ds4}r_{ds2}$
- $V_{MIN}(out) = 2V_{ON}$
- $V_{MIN}(in) = V_T + 2V_{ON}$
- Current gain matching is excellent



Self-biased, cascode current mirror

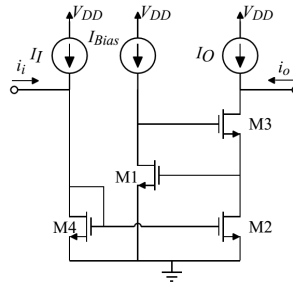


Small-signal model to calculate  $R_{in}$ .

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## MOSFET CURRENT MIRRORS

### Improved Current Mirrors – MOS Regulated Cascode Current Mirror



- Output resistance  $R_{out} \approx g_m^2 r_{ds}^3$
- Input resistance  $R_{in} \approx \frac{1}{g_{m4}}$
- $V_{MIN}(out) = V_T + 2V_{ON}$  (Can be reduced to  $2V_{ON}$ )
- $V_{MIN}(in) = V_T + V_{ON}$  (Can be reduced to  $V_{ON}$ )
- Current gain matching - good as long as  $V_{DS4} = V_{DS2}$

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## MOSFET CURRENT MIRRORS

### Summary of MOS Current Mirrors

| Current Mirror            | Accuracy       | Output Resistance | Input Resistance    | Minimum Output Voltage                 | Minimum Input Voltage                        |
|---------------------------|----------------|-------------------|---------------------|----------------------------------------|----------------------------------------------|
| Simple                    | Poor           | $r_{ds}$          | $\frac{1}{g_m}$     | $V_{ON}$                               | $V_T + V_{ON}$                               |
| Wide Output Swing Cascode | Excellent      | $g_m r_{ds}^2$    | $\frac{1}{g_m}$     | $2V_{ON}$                              | $V_T + V_{ON}$                               |
| Self-biased Cascode       | Excellent      | $g_m r_{ds}^2$    | $R + \frac{1}{g_m}$ | $2V_{ON}$                              | $V_T + 2V_{ON}$                              |
| Regulated Cascode         | Good-Excellent | $g_m^2 r_{ds}^3$  | $\frac{1}{g_m}$     | $V_T + 2V_{ON}$<br>(Can be $2V_{ON}$ ) | $V_T + V_{ON}$<br>(Can be $\approx V_{ON}$ ) |

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## INVERTING AMPLIFIERS

### Types of Amplifiers

| Type of Amplifier | Gain = $\frac{\text{Output}}{\text{Input}}$                | Ideal Input Resistance | Ideal Output Resistance |
|-------------------|------------------------------------------------------------|------------------------|-------------------------|
| Voltage           | $A_v = \frac{\text{Output Voltage}}{\text{Input Voltage}}$ | Infinite               | Zero                    |
| Current           | $A_i = \frac{\text{Output Current}}{\text{Input Current}}$ | Zero                   | Infinite                |
| Transconductance  | $G_m = \frac{\text{Output Current}}{\text{Input Voltage}}$ | Infinite               | Infinite                |
| Transresistance   | $R_m = \frac{\text{Output Voltage}}{\text{Input Current}}$ | Zero                   | Zero                    |

- Most CMOS amplifiers fit naturally into the transconductance amplifier category as they have large input resistance and fairly large output resistance.
- If the load resistance is high, the CMOS transconductance amplifier is essentially a voltage amplifier.

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## INVERTING AMPLIFIERS

### Characterization of an Amplifier

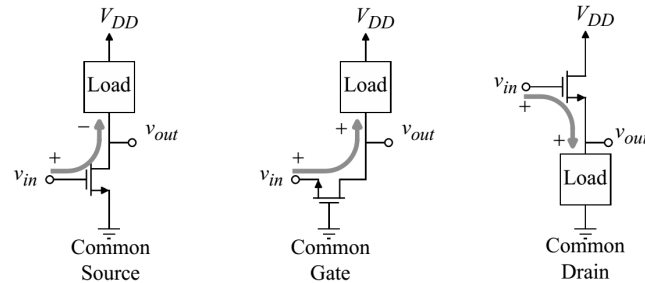
1. Large signal static characterization:
  - Plot of output versus input (transfer curve)
  - Large signal gain
  - Output and input swing limits
2. Small signal static characterization:
  - AC gain
  - AC input resistance
  - AC output resistance
3. Small signal dynamic characterization:
  - Bandwidth
  - Noise
  - Power supply rejection
4. Large signal dynamic characterization:
  - Slew rate
  - Nonlinearity

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## INVERTING AMPLIFIERS

### Inverting and Noninverting Amplifiers

- The types of amplifiers are based on the various configurations of the actual transistors.
- If we assume that one terminal of the transistor is grounded, then three possibilities result:

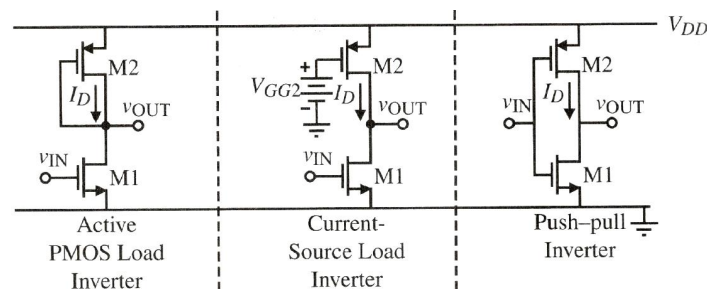


- Note that there are two categories of amplifiers:
  - Noninverting - Those whose input and output are in phase (common gate and common drain)
  - Inverting - Those whose input and output are out of phase (common source)

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## INVERTING AMPLIFIERS

- Typically, the inverter uses the common source configuration with either an active resistor for a load or a current sink/source as a load resistor.
- These inverters include the active PMOS load inverter, current-source load inverter, push-pull inverter.

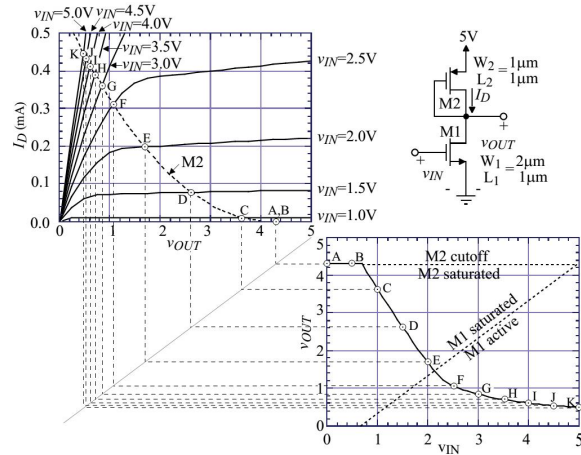


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## ACTIVE LOAD INVERTING AMPLIFIERS

### Voltage Transfer Characteristic of the Active Load Inverter

- The large-signal characteristics
- Consider large-signal swing limitations of the active load inverter
  - For maximum output voltage,  $v_{out(max)}$ : Assuming M1 is off,  $v_{out(max)} \approx V_{DD} - |V_{TP}|$  (This limits the subthreshold current flowing in MOSFET. This very small current will eventually allow the output voltage to approach  $V_{DD}$ .)
  - For minimum output voltage,  $v_{out(min)}$ : Assuming M1 is in non-saturated (active) region and  $V_{T1} = |V_{T2}| = V_T$ .
  - The boundary between active and saturation operation for M1 is  $v_{DS1} \geq v_{GS1} - V_{TN} \rightarrow v_{OUT} \geq v_{IN} - 0.7V$

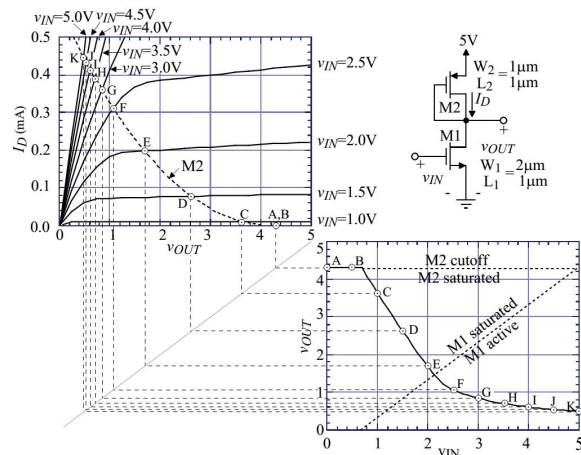


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## ACTIVE LOAD INVERTING AMPLIFIERS

### Voltage Transfer Characteristic of the Active Load Inverter

- $v_{in}$  vs.  $v_{out}$  curve is not good
- The output voltage can not go to the lower limit (0V): the voltage across M2 produces current that must flow through M1
- Any MOSFET can only have zero voltage across its drain-source if  $i_D = 0$ . The minimum value of  $v_{out}$  is equal to whatever drain-source drop across M1 is required to support the current defined by M2.



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## ACTIVE LOAD INVERTING AMPLIFIERS

### Large-Signal Voltage Swing Limits of the Active Load Inverter

- Maximum output voltage,  $v_{OUT}(\max)$ :  $v_{OUT}(\max) \cong V_{DD} - |V_{TP}|$   
(ignores subthreshold current influence on the MOSFET)

- Minimum output voltage,  $v_{OUT}(\min)$ :

Assume that M1 is nonsaturated and that  $V_{T1} = |V_{T2}| = V_T$ .

$$v_{DS1} \leq v_{GS1} - V_{TN} \rightarrow v_{OUT} \leq v_{IN} - 0.7V$$

- The current through M1 is

$$i_D = \beta_1 \left( (v_{GS1} - V_T) v_{DS1} - \frac{v_{DS1}^2}{2} \right) = \beta_1 \left( (V_{DD} - V_T)(v_{OUT}) - \frac{(v_{OUT})^2}{2} \right)$$

and the current through M2 is

$$i_D = \frac{\beta_2}{2} (v_{SG2} - V_T)^2 = \frac{\beta_2}{2} (V_{DD} - v_{OUT} - V_T)^2 = \frac{\beta_2}{2} (v_{OUT} + V_T - V_{DD})^2$$

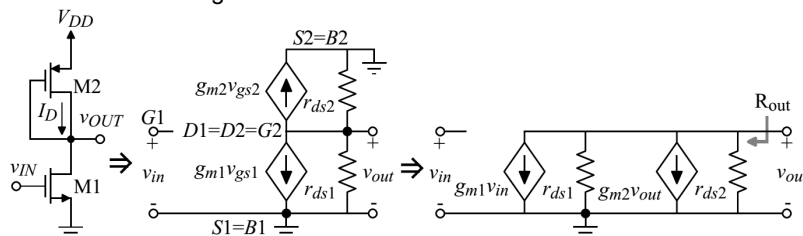
- Equating these currents gives the minimum  $v_{OUT}$  as:  $v_{OUT}(\min) = V_{DD} - V_T - \frac{V_{DD} - V_T}{\sqrt{1 + (\beta_2/\beta_1)}}$

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## ACTIVE LOAD INVERTING AMPLIFIERS

### Small-Signal Performance of the Active Load Inverter

- The development of the small-signal model for the active load inverter is shown below:



- Sum the currents at the output node to get:  $g_{m1}v_{in} + g_{ds1}v_{out} + g_{m2}v_{out} + g_{ds2}v_{out} = 0$
- Solving for the voltage gain,  $v_{out}/v_{in}$ , gives

$$\frac{v_{out}}{v_{in}} = \frac{-g_{m1}}{g_{ds1} + g_{ds2} + g_{m2}} \cong -\frac{g_{m1}}{g_{m2}} = -\left(\frac{K'_N W_1 L_2}{K'_P L_1 W_2}\right)^{1/2}$$

- The small-signal output resistance can also be found from the above by letting  $v_{in} = 0$  to get,

$$R_{out} = \frac{1}{g_{ds1} + g_{ds2} + g_{m2}} \cong \frac{1}{g_{m2}}$$

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## ACTIVE LOAD INVERTING AMPLIFIERS

### Frequency Response of the Active Load Inverter

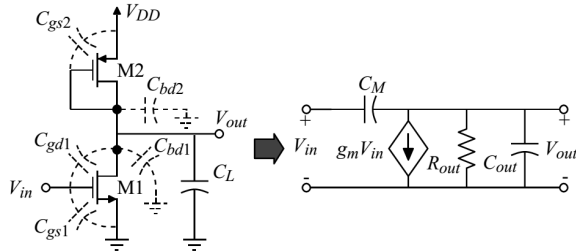
- Incorporation of the parasitic capacitors into the small-signal model:
- If we assume the input voltage has a small source resistance, then we can

write the following:  $sC_M(V_{out} - V_{in}) + g_m V_{in} + G_{out} V_{out} + sC_{out} V_{out} = 0$

$$\frac{V_{out}}{V_{in}} = \frac{-(g_m - sC_M)}{G_{out} + sC_M + sC_{out}} = -g_m R_{out} \left[ \frac{1 - \frac{sC_M}{g_m}}{1 + sR_{out}(C_M + C_{out})} \right] = \frac{-g_m R_{out} \left( 1 - \frac{s}{z_1} \right)}{1 - \frac{s}{p_1}}$$

where  $g_m = g_{m1}$ ,  $p_1 = -1/R_{out}(C_{out} + C_M)$ ,  $z_1 = g_{m1}/C_M$

and  $R_{out} = [g_{ds1} + g_{ds2} + g_{m2}]^{-1} \approx 1/g_{m2}$ ,  $C_M = C_{gd1}$ ,  $C_{out} = C_{bd1} + C_{bd2} + C_{bd1} + C_{gs2} + C_L$



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## ACTIVE LOAD INVERTING AMPLIFIERS

### Performance of an Active Load Inverter

- **Example 1:** Calculate the output-voltage swing limits for  $V_{DD} = 5$  volts, the small-signal gain, the output resistance, and the -3 dB frequency of active load inverter if  $(W_1/L_1)$  is  $2 \mu\text{m}/1 \mu\text{m}$  and  $W_2/L_2 = 1 \mu\text{m}/1 \mu\text{m}$ ,  $C_{gd1} = 100\text{fF}$ ,  $C_{bd1} = 200\text{fF}$ ,  $C_{bd2} = 100\text{fF}$ ,  $C_{gs2} = 200\text{fF}$ ,  $C_L = 1 \text{ pF}$ , and  $I_{D1} = I_{D2} = 100\mu\text{A}$ , using the parameters given in the previous slides.
- **Solution:** From the above results we find that:
  - $v_{OUT}(\text{max}) = 4.3$  volts
  - $v_{OUT}(\text{min}) = 0.418$  volts
  - Small-signal voltage gain =  $-1.92\text{V/V}$
  - $R_{out} = 9.17 \text{ k}\Omega$  including  $g_{ds1}$  and  $g_{ds2}$  and  $10 \text{ k}\Omega$  ignoring  $g_{ds1}$  and  $g_{ds2}$
  - $z_1 = 2.10 \times 10^9 \text{ rads/sec}$
  - $p_1 = -68.127 \times 10^6 \text{ rads/sec}$ .
  - Thus, the -3 dB frequency is  $10.84 \text{ MHz}$ .

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## CURRENT SOURCE INVERTER

### Voltage Transfer Characteristic of the Current Source Inverter

- The current-source inverter has gain higher than the active load inverter.
- $v_{in}$  vs.  $v_{out}$  curve is better than the active load inverter.
- A current source load is used instead of a PMOS as the load. The current source is a common gate configuration using PMOS with the gate connected to a DC bias voltage,  $V_{GG}$ .
- The limits of the large-signal output-voltage swing can be found by the following approach:
  - The upper limit (headroom)  $v_{out(max)}$ : equal to  $V_{DD}$  since M1 is off, the voltage across M2 can go to zero, allowing the output voltage to equal  $V_{DD}$  providing no output DC current is required  $\rightarrow v_{out(max)} \approx V_{DD}$ .
  - The lower limit (legroom)  $v_{out(min)}$ : assuming that M1 will be in non-saturation region.

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## CURRENT SOURCE INVERTER

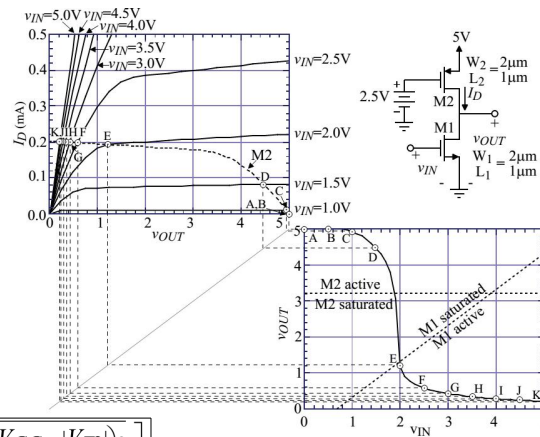
### Voltage Transfer Characteristic of the Current Source Inverter

#### Regions of operation for the transistors:

- M1:  $v_{DS1} \geq v_{GS1} - V_{Tn}$   
or  $v_{OUT} \geq v_{IN} - 0.7V$
- M2:  $v_{SD2} \geq v_{SG2} - |V_{Tp}|$   
or  $V_{DD} - v_{OUT} \geq V_{DD} - V_{GG2} - |V_{Tp}|$   
or  $v_{OUT} \leq 3.2V$
- Swing limits:

$$v_{OUT(max)} \approx V_{DD}$$

$$v_{OUT(min)} = (V_{DD} - V_{T1}) \left[ 1 - \sqrt{1 - \left( \frac{\beta_2}{\beta_1} \right) \left( \frac{V_{DD} - V_{GG} - |V_{T2}|}{V_{DD} - V_{T1}} \right)^2} \right]$$

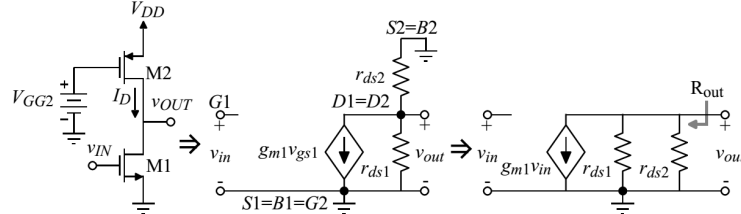


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## CURRENT SOURCE INVERTER

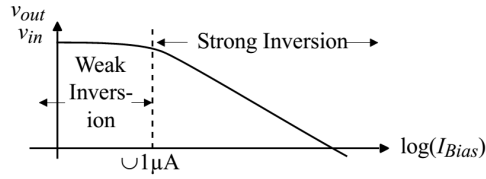
### Small-Signal Performance of the Current Source Load Inverter

- Small-Signal Model:



- Midband Performance:

$$\frac{v_{out}}{v_{in}} = \frac{-g_{m1}}{g_{ds1} + g_{ds2}} = \left( \frac{2K'_N W_1}{L_1 I_D} \right)^{1/2} \left( \frac{-1}{\lambda_1 + \lambda_2} \right) \propto \frac{1}{\sqrt{I_D}} !!! \quad \text{and} \quad R_{out} = \frac{1}{g_{ds1} + g_{ds2}} \cong \frac{1}{I_D(\lambda_1 + \lambda_2)}$$



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## CURRENT SOURCE INVERTER

### Frequency Response of the Current Source Load Inverter

- Incorporation of the parasitic capacitors into the small-signal model (x is connected to  $V_{GG2}$ ):
- If we assume the input voltage has a small source resistance, then we can write the following:

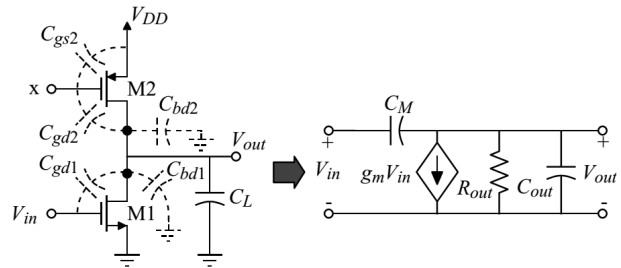
$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{-g_m R_{out} \left( 1 - \frac{s}{z_1} \right)}{1 - \frac{s}{p_1}}$$

where  $g_m = g_{m1}$ ,  $p_1 = -1/R_{out}(C_{out} + C_M)$ ,  $z_1 = g_m/C_M$

and  $R_{out} = 1/(g_{ds1} + g_{ds2})$ ,  $C_M = C_{gd1}$ ,  $C_{out} = C_{gd2} + C_{bd1} + C_{bd2} + C_L$

Therefore, if  $|p_1| < |z_1|$ , then the -3 dB frequency response can be expressed as

$$\omega_{-3dB} \approx \omega_1 = \frac{g_{ds1} + g_{ds2}}{C_{gd1} + C_{gd2} + C_{bd1} + C_{bd2} + C_L}$$



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## CURRENT SOURCE INVERTER

### Performance of the Current Source Load Inverter

- **Example 2:** A current-sink inverter is shown. Assume that  $W_1 = 2\mu\text{m}$ ,  $L_1 = 1\mu\text{m}$ ,  $W_2 = 1\mu\text{m}$ ,  $L_2 = 1\mu\text{m}$ ,  $V_{DD} = 5\text{ volts}$ ,  $V_{NB1} = 3\text{ volts}$ , and the parameters of Table 3.1 -2 describe M1 and M2. Use the capacitor values of Example 1 ( $C_{gd1} = C_{gd2}$ ). Calculate the output-swing limits and the small-signal performance.
- **Solution:** To attain the output signal-swing limitations, treat the current sink inverter as a current source CMOS inverter with PMOS (NMOS) parameters for the NMOS (PMOS) and use NMOS equations. Using a prime notation to designate the results of the current source CMOS inverter that exchanges the PMOS and NMOS model parameters,

$$v_{OUT}(\text{max})' = 5\text{V and } v_{OUT}(\text{min})' = (5-0.7) \left[ 1 - \sqrt{1 - \left( \frac{110 \cdot 1}{50 \cdot 2} \left( \frac{3-0.7}{5-0.7} \right)^2} \right)} \right] = 0.74\text{V}$$

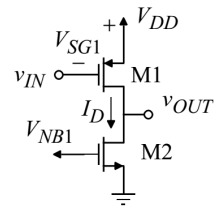
- In terms of the current sink CMOS inverter, these limits are subtracted from 5V to get

$$v_{OUT}(\text{max}) = 4.26\text{V and } v_{OUT}(\text{min}) = 0\text{V.}$$

- To find the small signal performance, first calculate the dc current. The dc current,  $I_D$ , is

$$I_D = \frac{K_N' W_1}{2L_1} (V_{GG1} - V_{TN})^2 = \frac{110 \cdot 1}{2 \cdot 1} (3-0.7)^2 = 291\mu\text{A}$$

$$v_{out}/v_{in} = -9.2\text{ V/V, } R_{out} = 38.1\text{ k}\Omega, f_{-3\text{dB}} = 2.78\text{ MHz}$$



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## PUSH-PULL INVERTING AMPLIFIER

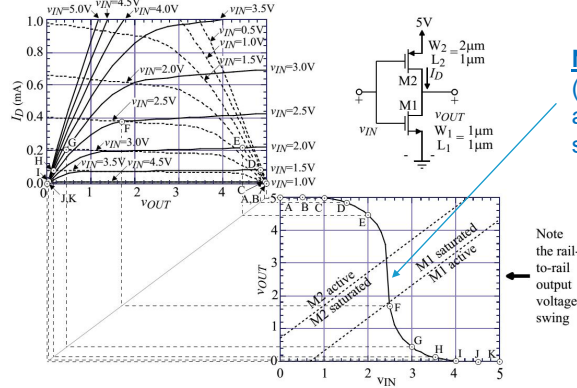
### Voltage Transfer Characteristic of the Push-Pull Inverting Amplifier

- The push-pull inverter has gain higher than the active load inverter and the current-source inverter (assuming identical transistors).
- Due to the fact that both transistors are being given by  $v_{in}$ .
- The output swing is capable of operation from rail-to-rail ( $V_{DD}$  to 0V)

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## PUSH-PULL INVERTING AMPLIFIER

### Voltage Transfer Characteristic of the Push-Pull Inverting Amplifier

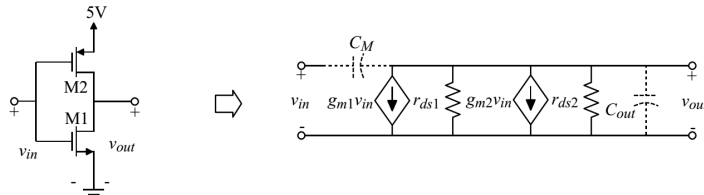


- Regions of operation for M1 and M2:
  - M1 (saturation):  $v_{DS1} \geq v_{GS1} - V_{T1} \rightarrow v_{OUT} \geq v_{IN} - 0.7V$
  - M2 (saturation):  $v_{SD2} \geq v_{SG2} - |V_{T2}| \rightarrow v_{DD} - v_{OUT} \geq v_{DD} - v_{IN} - |V_{T2}| \rightarrow v_{OUT} \leq v_{IN} + 0.7V$

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## PUSH-PULL INVERTING AMPLIFIER

### Small-Signal Performance of the Push-Pull Amplifier



- Small-signal analysis gives the following results:

$$\frac{v_{out}}{v_{in}} = \frac{-(g_{m1} + g_{m2})}{g_{ds1} + g_{ds2}} = -\sqrt{(2/I_D)} \left[ \frac{\sqrt{K'_N(W_1/L_1)} + \sqrt{K'_P(W_2/L_2)}}{\lambda_1 + \lambda_2} \right] \quad R_{out} = \frac{1}{g_{ds1} + g_{ds2}}$$

$$z = \frac{g_{m1} + g_{m2}}{C_M} = \frac{g_{m1} + g_{m2}}{C_{gd1} + C_{gd2}} \quad p_1 = \frac{-(g_{ds1} + g_{ds2})}{C_{gd1} + C_{gd2} + C_{bd1} + C_{bd2} + C_L}$$

- If  $z_1 > |p_1|$ , then

$$\omega_{3dB} = \frac{g_{ds1} + g_{ds2}}{C_{gd1} + C_{gd2} + C_{bd1} + C_{bd2} + C_L}$$

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## PUSH-PULL INVERTING AMPLIFIER

### Performance of the Push-Pull Amplifier

- **Example 3:** The performance of a push-pull CMOS inverter is to be examined. Assume that  $W_1 = 1 \mu\text{m}$ ,  $L_1 = 1 \mu\text{m}$ ,  $W_2 = 2 \mu\text{m}$ ,  $L_2 = 1 \mu\text{m}$ ,  $V_{DD} = 5 \text{ volts}$ , and use the parameters of Table 3.1 - 2 to model M1 and M2. Use the capacitor values of Example 1 ( $C_{gd1} = C_{gd2}$ ). Calculate the output-swing limits and the small-signal performance assuming that  $I_{D1} = I_{D2} = 300 \mu\text{A}$ .
- **Solution:** The output swing is seen to be from 0V to 5V. In order to find the small signal performance, we will make the important assumption that both transistors are operating in the saturation region. Therefore:

$$\frac{v_{out}}{v_{in}} = \frac{-257 \mu\text{S} - 245 \mu\text{S}}{12 \mu\text{S} + 15 \mu\text{S}} = -18.6 \text{ V/V}$$

$$R_{out} = 37 \text{ k}\Omega$$

$$f_{-3\text{dB}} = 2.86 \text{ MHz}$$

$$z_1 = 399 \text{ MHz}$$

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## SUMMARY OF INVERTING AMPLIFIER

### Table of Performance

| Inverter                       | AC Voltage Gain                                | AC Output Resistance          | Bandwidth (CGB=0)                                                 | Equivalent, input-referred, mean-square noise voltage                                                             |
|--------------------------------|------------------------------------------------|-------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| p-channel active load inverter | $\frac{-g_{m1}}{g_{m2}}$                       | $\frac{1}{g_{m2}}$            | $\frac{g_{m2}}{C_{BD1} + C_{GS1} + C_{GS2} + C_{BD2}}$            | $e_{n1}^2 + e_{n2}^2 \left( \frac{g_{m2}}{g_{m1}} \right)^2$                                                      |
| Current source load inverter   | $\frac{-g_{m1}}{g_{ds1} + g_{ds2}}$            | $\frac{1}{g_{ds1} + g_{ds2}}$ | $\frac{g_{ds1} + g_{ds2}}{C_{BD1} + C_{GD1} + C_{DG2} + C_{BD2}}$ | $e_{n1}^2 + e_{n2}^2 \left( \frac{g_{m2}}{g_{m1}} \right)^2$                                                      |
| Push-Pull inverter             | $\frac{-(g_{m1} + g_{m2})}{g_{ds1} + g_{ds2}}$ | $\frac{1}{g_{ds1} + g_{ds2}}$ | $\frac{g_{ds1} + g_{ds2}}{C_{BD1} + C_{GD1} + C_{GS2} + C_{BD2}}$ | $\left( \frac{g_{m1} e_{n1}}{g_{m1} + g_{m2}} \right)^2 + \left( \frac{g_{m1} e_{n1}}{g_{m1} + g_{m2}} \right)^2$ |

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## DIFFERENTIAL AMPLIFIERS

### What is a Differential Amplifiers

- A differential amplifier is an amplifier that amplifies the difference between two voltages and rejects the average or common mode value of the two voltages.

- Differential and common mode voltages:

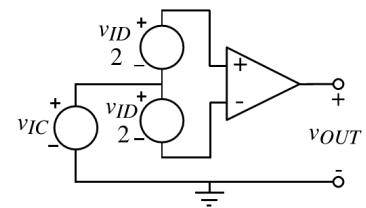
- $v_1$  and  $v_2$  are called single-ended voltages. They are voltages referenced to AC ground.
- The differential-mode input voltage,  $v_{ID}$ , is the voltage difference between  $v_1$  and  $v_2$ .
- The common-mode input voltage,  $v_{IC}$ , is the average value of  $v_1$  and  $v_2$ .

$$v_{ID} = v_1 - v_2, v_{IC} = (v_1 + v_2)/2 \rightarrow v_1 = v_{IC} + 0.5v_{ID} \text{ and } v_2 = v_{IC} - 0.5v_{ID}$$

$$v_{out} = A_{VD}v_{ID} \pm A_{VC}v_{IC} = A_{VD}(v_1 - v_2) \pm A_{VC}\left(\frac{v_1 + v_2}{2}\right)$$

where  $A_{VD}$  = differential-mode voltage gain

$A_{VC}$  = common-mode voltage gain

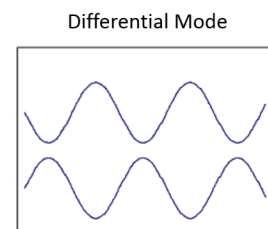
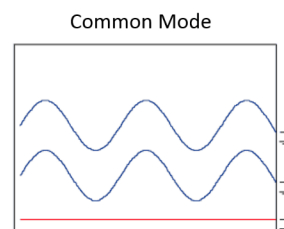
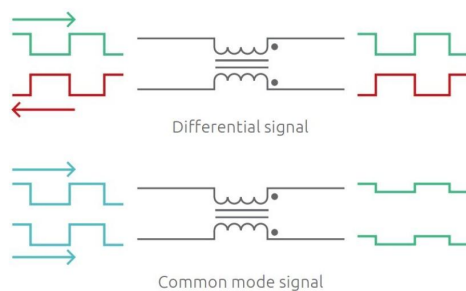


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## DIFFERENTIAL AMPLIFIERS

### Differential Amplifier Definitions

- Common-mode signal:** Common-mode signal is the voltage common to both input terminals of an electrical device.



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## DIFFERENTIAL AMPLIFIERS

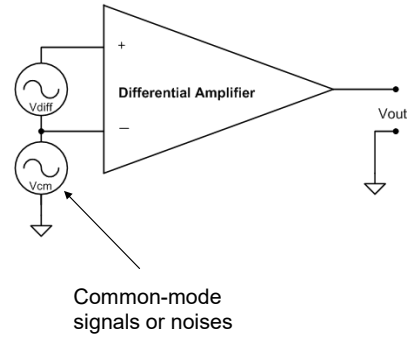
### Differential Amplifier Definitions

- **Common mode rejection ratio (CMRR):** CMRR is a ratio of differential gain to the common-mode gain, a metric used to quantify the ability of the device to reject common-mode signals (i.e. those that appear simultaneously and in-phase on both inputs.).

$$CMRR = \left| \frac{A_{VD}}{A_{VC}} \right| = 20 \log_{10} \left( \frac{A_{VD}}{A_{VC}} \right) (dB)$$

Ideally:  $V_o = A_{vd}(V_+ - V_-)$

In fact:  $V_o = A_{vd}(V_+ - V_-) + \frac{1}{2} A_{vc}(V_+ - V_-)$



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## DIFFERENTIAL AMPLIFIERS

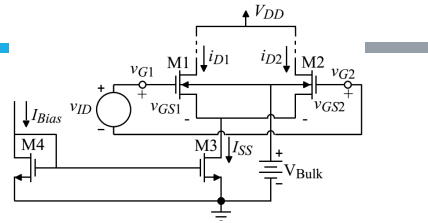
### Differential Amplifier Definitions

- **Input common-mode range (ICMR):** The input common-mode range is the range of common-mode voltages over which the differential amplifier continues to sense and amplify the difference signal with the same gain. Typically, the ICMR is defined by the common-mode voltage range over which all MOSFETs remain in the saturation region.
- **Input common-mode range (ICMR):** is also known as **input voltage swing** ( $V_{IC(min)} \rightarrow V_{IC(max)}$ ), the common mode input voltage range is always defined relative to **the positive supply and the negative supply**.
- **Output offset voltage ( $V_{OS(out)}$ ):** The output offset voltage is the voltage which appears at the output of the differential amplifier when the input terminals are connected together.
- **Input offset voltage ( $V_{OS(in)} = V_{OS}$ ):** The input offset voltage is equal to the output offset voltage divided by the differential voltage gain.

$$V_{OS(in)} = \left| \frac{V_{OS(out)}}{A_{VD}} \right|$$

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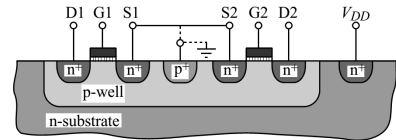
## DIFFERENTIAL AMPLIFIERS



### Large-signal Analysis of the Differential Amplifier

- M1 and M2 are perfectly matched. M1 and M2 are biased with a current sink  $I_{SS}$  connected to the sources of M1 and M2 → source-coupled pair.
- M3 and M4 are a case of the current sink  $I_{SS}$ .
- M1 and M2 are not connected to ground

→ where to connect the bulk pin ? → Answer: depending on technology.



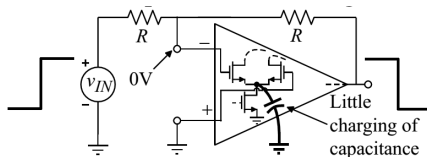
- If CMOS technology is p-well, NMOS transistors are fabricated on p-well
  - Connect the bulks to the sources of M1 and M2 and let p-well containing M1 and M2 float → threshold voltage will not increase because of the reverse-biased bulk-source junction. The capacitance from the source-coupled point to ground becomes the entire reverse-biased pn junctions between the p-well and the n-substrate.
  - Connect the bulks of M1 and M2 to ground → threshold voltage will increase and vary with common-mode input voltage, but the capacitance from the source-coupled point to ground is reduced to the two reverse-biased pn junctions between the sources of M1/M2 and p-well.

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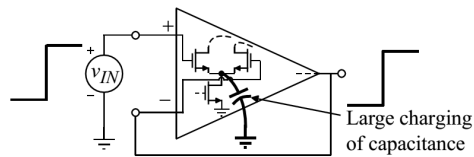
## DIFFERENTIAL AMPLIFIERS

### Large-Signal Analysis of the Differential Amplifier

- In summary:
  - Bulks connected to the sources: No modulation of  $V_T$  but large common mode parasitic capacitance.
  - Bulks connected to ground: Smaller common mode parasitic capacitors, but modulation of  $V_T$ .



Inverting configuration of the opamp with a gain of -1. The source-coupled node has little change in voltage to ground when 1V step is applied at input.



Noninverting configuration of the opamp with a gain of +1. The source-coupled node changes the same amount as the 1V input step.

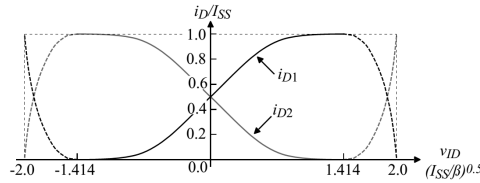
- Normally, the input to an OP-AMP is a differential amplifier.

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## DIFFERENTIAL AMPLIFIERS

### Transconductance Characteristic of the Differential Amplifier

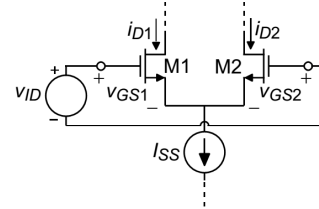
- Defining equations:  $v_{ID} = v_{GS1} - v_{GS2} = \sqrt{\frac{2i_{D1}}{\beta}} - \sqrt{\frac{2i_{D2}}{\beta}}$  and  $I_{SS} = i_{D1} + i_{D2}$
- Solutions:  $i_{D1} = \frac{I_{SS}}{2} + \frac{I_{SS}}{2} \left( \frac{\beta v_{ID}^2}{I_{SS}} - \frac{\beta^2 v_{ID}^4}{4I_{SS}^2} \right)^{1/2}$  and  $i_{D2} = \frac{I_{SS}}{2} - \frac{I_{SS}}{2} \left( \frac{\beta v_{ID}^2}{I_{SS}} - \frac{\beta^2 v_{ID}^4}{4I_{SS}^2} \right)^{1/2}$   
which are valid for  $v_{ID} < 2(I_{SS}/\beta)^{1/2}$
- Plot of the normalized  $i_D$  of M1 vs. the normalized differential input voltage:



- Differentiating  $i_{D1}$  (or  $i_{D2}$ ) with respect to  $v_{ID}$  and setting  $V_{ID} = 0V$  gives

$$g_m = \frac{di_{D1}}{dv_{ID}}(V_{ID} = 0) = \sqrt{\frac{\beta I_{SS}}{4}} = \sqrt{\frac{K_1 I_{SS} W_1}{4L_1}} \quad (\text{half the } g_m \text{ of an inverting amplifier})$$

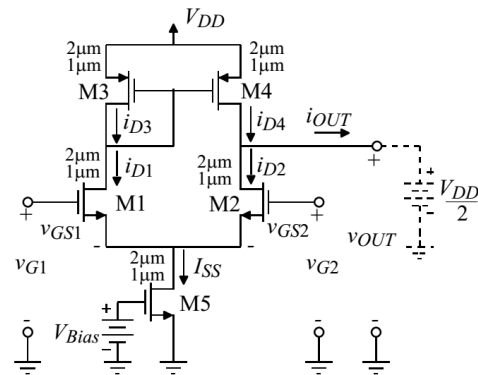
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## DIFFERENTIAL AMPLIFIER WITH A CURRENT MIRROR LOAD

### Operations of the Differential Amplifier

- Under quiescent conditions (no applied differential signal), the two currents of M1 and M2 are equal and sum to  $I_{SS}$ , the current in current sink M5.
- The current of M1 will determine the current in M3, this current will be mirrored in M4. If  $v_{GS1} = v_{GS2}$ , M1 and M2 are matched, then the currents of M1 and M2 are equal.
- Thus, current that M4 sources to M2 should be equal to the current that M2 requires, causing  $i_{OUT}$  to be zero.
- All transistors are assumed to be saturated.
- If  $v_{GS1} > v_{GS2}$ , then  $i_{D1}$  increases with respect to  $i_{D2}$  since  $I_{SS} = i_{D1} + i_{D2} \rightarrow$  imply an increase in  $i_{D3} + i_{D4}$ , but  $i_{D2}$  decreases.
- To make the circuit equilibrium:  $i_{OUT}$  becomes positive and  $v_{OUT}$  to increase.
- If  $v_{GS1} < v_{GS2}$ ,  $i_{OUT}$  becomes negative and  $v_{OUT}$  decreases.



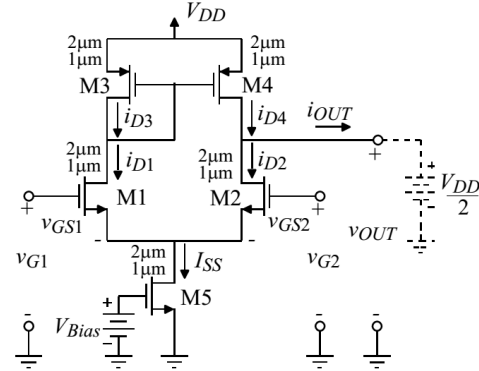
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## DIFFERENTIAL AMPLIFIER WITH A CURRENT MIRROR LOAD

### Voltage Transfer Characteristic of the Differential Amplifier

- In order to obtain the voltage transfer characteristic, a load for the differential amplifier must be defined. We will select a current mirror load as illustrated below.
- Note that output signal to ground is equivalent to the differential output signal due to the current mirror.
- The short-circuit, transconductance is given as

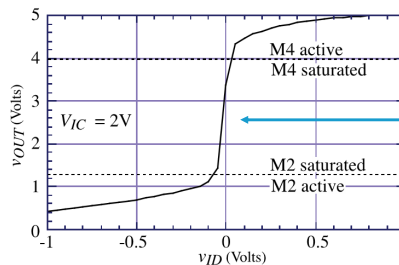
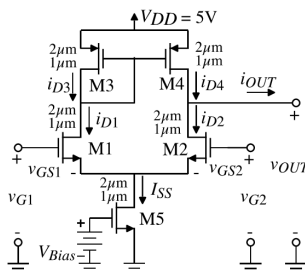
$$g_m = \frac{di_{OUT}}{dv_{ID}} (V_{ID} = 0) = \sqrt{\beta I_{SS}} = \sqrt{\frac{K'_1 I_{SS} W_1}{L_1}}$$



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## DIFFERENTIAL AMPLIFIERS

### Voltage Transfer Function of the Differential Amplifier with a Current Mirror Load



The largest gain  
M2/M4 saturated

- Regions of operation of the transistors:
  - M2 is saturated when,  $v_{DS2} \geq v_{GS2} - V_{TN} \rightarrow v_{OUT} - V_{S1} \geq V_{IC} - 0.5v_{ID} - V_{S1} - V_{TN} \rightarrow v_{OUT} \geq V_{IC} - V_{TN}$   
where we have assumed that the region of transition for M2 is close to  $v_{ID} = 0V$ .
  - M4 is saturated when,  $v_{SD4} \geq v_{SG4} - |V_{TP}| \rightarrow V_{DD} - v_{OUT} \geq V_{SG4} - |V_{TP}| \rightarrow v_{OUT} \leq V_{DD} - V_{SG4} + |V_{TP}|$
- The regions of operations shown on the voltage transfer function assume  $I_{SS} = 100\mu A$ .

$$V_{SG4} = \sqrt{\frac{2 \cdot 50}{50 \cdot 2}} + |V_{TP}| = 1 + |V_{TP}| \Rightarrow v_{OUT} \leq 5 - 1 - 0.7 + 0.7 = 4V$$

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## DIFFERENTIAL AMPLIFIERS

### Input Common Mode Range (ICMR)

- ICMR is found by setting  $v_{ID} = 0$  and varying  $v_{IC}$  until one of the transistors leaves the saturation.

### Highest Common Mode Voltage

- Path from G1 through M1 and M3 to  $V_{DD}$ :

$$V_{IC}(\max) = V_{G1}(\max) = V_{G2}(\max) = V_{DD} - V_{SG3} - V_{DS1}(\text{sat}) + V_{GS1}$$

$$\text{or } V_{IC}(\max) = V_{DD} - V_{SG3} + V_{TN1}$$

- Path from G2 through M2 and M4 to  $V_{DD}$ :

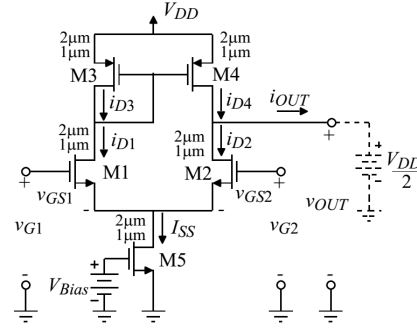
$$V_{IC}(\max)' = V_{DD} - V_{SD4}(\text{sat}) - V_{DS2}(\text{sat}) + V_{GS2} = V_{DD} - V_{SD4}(\text{sat}) + V_{TN2}$$

$$V_{IC}(\max) = V_{DD} - V_{SG3} + V_{TN1}$$

### Lowest Common Mode Voltage (Assume a $V_{SS}$ for generality)

$$V_{IC}(\min) = V_{SS} + V_{DS5}(\text{sat}) + V_{GS1} = V_{SS} + V_{DS5}(\text{sat}) + V_{GS2}$$

where we have assumed that  $V_{GS1} = V_{GS2}$  during changes in the input common mode voltage.

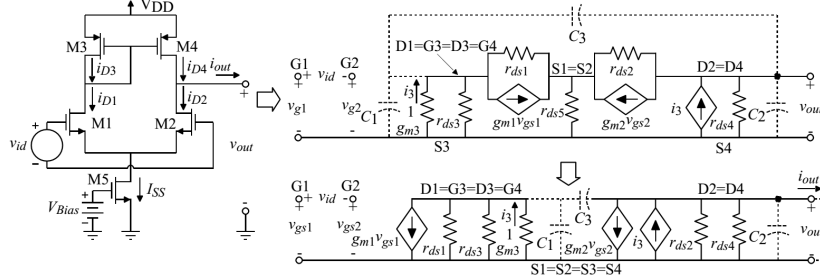


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## DIFFERENTIAL AMPLIFIERS

### Small-Signal Analysis of the Differential-Mode of the Diff. Amp

- A requirement for differential-mode operation is that the differential amplifier is balanced



### Differential Transconductance:

- Assume that the output of the differential amplifier is an AC short.

$$i_{out}' = \frac{g_{m1}g_{m3}r_{p1}}{1 + g_{m3}r_{p1}} v_{gs1} - g_{m2}v_{gs2} \approx g_{m1}v_{gs1} - g_{m2}v_{gs2} = g_{md}v_{id}$$

where  $g_{m1} = g_{m2} = g_{md}$ ,  $r_{p1} = r_{ds1} || r_{ds3}$  and  $i_{out}'$  designates the output current into a short circuit.

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## DIFFERENTIAL AMPLIFIERS

### Small-Signal Analysis of the Differential-Mode of the Diff. Amp

- Output resistance

$$r_{out} = \frac{1}{g_{ds2} + g_{ds4}} = r_{ds2} || r_{ds4}$$

- Differential Voltage Gain

$$A_v = \frac{v_{out}}{v_{id}} = \frac{g_{md}}{g_{ds2} + g_{ds4}}$$

- If we assume that all transistors are in saturation and replace the small signal parameters of  $g_m$  and  $r_{ds}$  in terms of their large-signal model equivalents, we achieve

$$A_v = \frac{v_{out}}{v_{id}} = \frac{(K_1 I_{SS} W_1 / L_1)^{1/2}}{(\lambda_2 + \lambda_4)(I_{SS}/2)} = \frac{2}{\lambda_2 + \lambda_4} \left( \frac{K_1 W_1}{I_{SS} L_1} \right)^{1/2} \propto \frac{1}{\sqrt{I_{SS}}}$$

- Note that the small-signal gain is inversely proportional to the square root of the bias current!

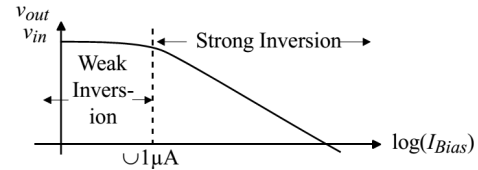
- Example:

- If  $W_1/L_1 = 2\mu\text{m}/1\mu\text{m}$  and  $I_{SS} = 50\mu\text{A}$  ( $10\mu\text{A}$ ), then

- $A_v(\text{n-channel}) = 46.6\text{V/V}$  ( $104.23\text{V/V}$ )

- $A_v(\text{p-channel}) = 31.4\text{V/V}$  ( $70.27\text{V/V}$ )

$$r_{out} = \frac{1}{g_{ds2} + g_{ds4}} = \frac{1}{25\mu\text{A} \cdot 0.09\text{V}^{-1}} = 0.444\text{M}\Omega \text{ (2.22M}\Omega\text{)}$$

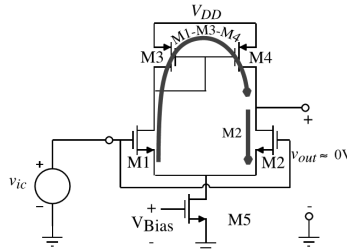


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## DIFFERENTIAL AMPLIFIERS

### Common Mode Analysis for the Current Mirror Load Differential Amplifier

- The current mirror load differential amplifier is not a good example for common mode analysis because the current mirror rejects the common mode signal.



$$\begin{bmatrix} \text{Total common mode Output due to } v_{ic} \end{bmatrix} = \begin{bmatrix} \text{Common mode output due to M1-M3-M4 path} \end{bmatrix} - \begin{bmatrix} \text{Common mode output due to M2 path} \end{bmatrix}$$

- Therefore:

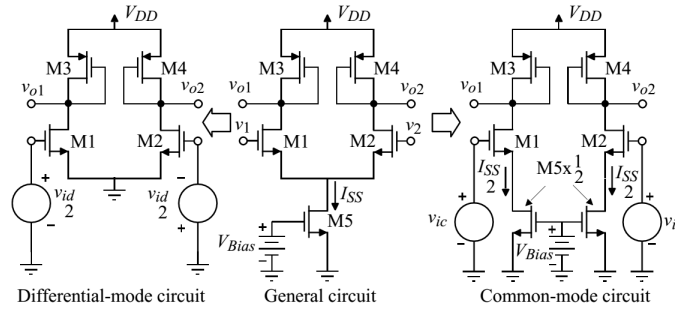
- The common mode output voltage should ideally be zero.
- Any voltage that exists at the output is due to mismatches in the gain between the two different paths.

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## DIFFERENTIAL AMPLIFIERS

### Small-Signal Analysis of the Common-Mode of the Differential Amplifier

- The common-mode gain of the differential amplifier with a current mirror load is ideally zero. To illustrate the common-mode gain, we need a different type of load so we will consider the following:



- Differential-Mode Analysis:

$$\frac{v_{o1}}{v_{id}} \approx -\frac{g_{m1}}{2g_{m3}} \quad \frac{v_{o2}}{v_{id}} \approx +\frac{g_{m2}}{2g_{m4}}$$

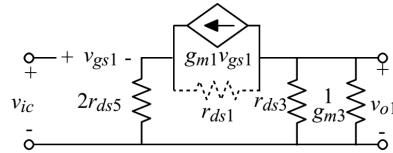
- Note that these voltage gains are half of the active load inverter voltage gain.

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## DIFFERENTIAL AMPLIFIERS

### Small-Signal Analysis of the Common-Mode of the Differential Amplifier

- Common-Mode Analysis:



- Assume that  $r_{ds1}$  is large and can be ignored (greatly simplifies the analysis):  $v_{gs1} = v_{g1} - v_{s1} = v_{ic} - 2g_{m1}r_{ds5}v_{gs1}$

- Solving for  $v_{gs1}$  gives

$$v_{gs1} = \frac{v_{ic}}{1 + 2g_{m1}r_{ds5}}$$

- The single-ended output voltage,  $v_{o1}$ , as a function of  $v_{ic}$  can be written as

$$\frac{v_{o1}}{v_{ic}} = -\frac{g_{m1}[r_{ds3}](1/g_{m3})}{1 + 2g_{m1}r_{ds5}} \approx -\frac{(g_{m1}/g_{m3})}{1 + 2g_{m1}r_{ds5}} \approx -\frac{g_{ds5}}{2g_{m3}}$$

- Common-Mode Rejection Ratio (CMRR):

$$CMRR = \frac{|v_{o1}/v_{id}|}{|v_{o1}/v_{ic}|} = \frac{g_{m1}/2g_{m3}}{g_{ds5}/2g_{m3}} = g_{m1}r_{ds5}$$

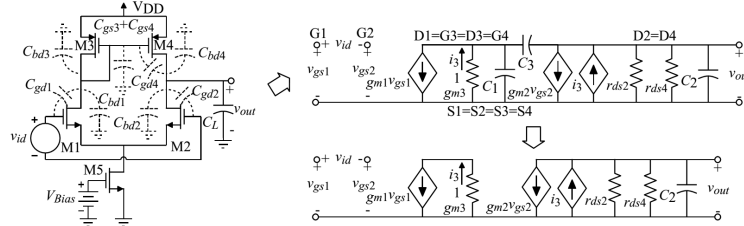
- How could you easily increase the CMRR of this differential amplifier?

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## DIFFERENTIAL AMPLIFIERS

### Frequency Response of the Differential Amplifier

- Back to the current mirror load differential amplifier:



- Ignore the zeros that occur due to  $C_{gd1}$ ,  $C_{gd2}$  and  $C_{gd4}$ .

$$C_1 = C_{gd1} + C_{bd1} + C_{bd3} + C_{gs3} + C_{gs4}, C_2 = C_{bd2} + C_{bd4} + C_{gd2} + C_L \text{ and } C_3 = C_{gd4}$$

- The poles are  $p_1 = -g_{m3}/C_1$  and  $p_2 = -(g_{ds2} + g_{ds4})/C_2$ . Since  $|p_1| \gg |p_2|$ , then we can write

$$V_{out}(s) \approx \frac{g_{m1}}{g_{ds2} + g_{ds4}} \left( \frac{\omega_2}{s + \omega_2} \right) [V_{gs1}(s) - V_{gs2}(s)] \quad \omega_2 \approx \frac{g_{ds2} + g_{ds4}}{C_2}$$

- The approximate frequency response of the differential amplifier reduces to  $\frac{V_{out}(s)}{V_{id}(s)} \approx \left( \frac{g_{m1}}{g_{ds2} + g_{ds4}} \right) \left( \frac{\omega_2}{s + \omega_2} \right)$

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## SMALL SIGNAL PERFORMANCE OF THE DIFFERENTIAL AMPLIFIER

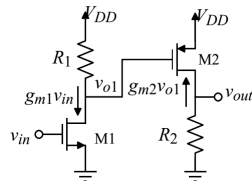
### Simplification of Small Signal Analysis

- Small signal analysis is used so often in analog circuit design that it becomes desirable to find faster ways of performing this important analysis.

### Intuitive Analysis (or Schematic Analysis)

- Technique:
  - Identify the transistor(s) that convert the input voltage to current (these transistors are called transconductance transistors).
  - Trace the currents to where they flow into an equivalent resistance to ground.
  - Multiply this resistance by the current to get the voltage at this node to ground.
  - Repeat this process until the output is reached.

- Simple Example:



$$v_{o1} = -(g_{m1}v_{in})R_1 \rightarrow v_{out} = -(g_{m2}v_{o1})R_2 \rightarrow v_{out} = (g_{m1}R_1g_{m2}R_2)v_{in}$$

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## SMALL SIGNAL PERFORMANCE OF THE DIFFERENTIAL AMPLIFIER

### Intuitive Analysis of the Current-Mirror Load Differential Amplifier

1.  $i_1 = 0.5g_{m1}v_{id}$  and  $i_2 = -0.5g_{m2}v_{id}$
2.  $i_3 = i_1 = 0.5g_{m1}v_{id}$
3.  $i_4 = i_3 = 0.5g_{m1}v_{id}$
4. The short-circuit output current is

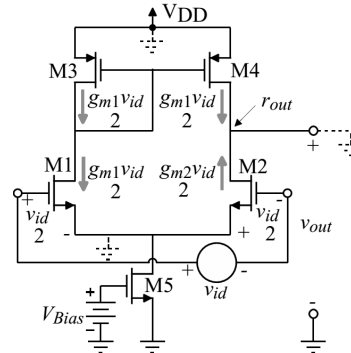
$$i_4 - i_2 = 0.5g_{m1}v_{id} + 0.5g_{m2}v_{id} = g_{m1}v_{id}$$

5. The resistance at the output node,  $r_{out}$ , is

$$r_{ds2} \parallel r_{ds4} \text{ or } \frac{1}{g_{ds2} + g_{ds4}}$$

6.  $v_{out} = (0.5g_{m1}v_{id} + 0.5g_{m2}v_{id})r_{out}$   

$$= \frac{g_{m1}v_{in}}{g_{ds2} + g_{ds4}} = \frac{g_{m2}v_{in}}{g_{ds2} + g_{ds4}} \Rightarrow \frac{v_{out}}{v_{in}} = \frac{g_{m1}}{g_{ds2} + g_{ds4}}$$



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## SMALL SIGNAL PERFORMANCE OF THE DIFFERENTIAL AMPLIFIER

### Some Concepts to Help Extend the Intuitive Method of Small-Signal Analysis

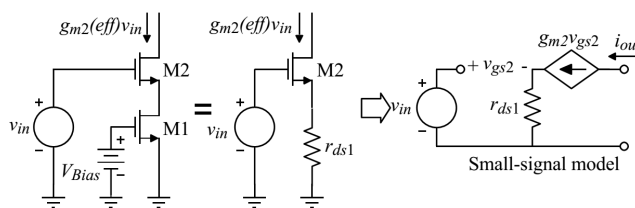
1. Approximate the output resistance of any cascode circuit as

$$R_{out} \approx (g_{m2}r_{ds2})r_{ds1}$$

where M1 is a transistor cascoded by M2.

2. If there is a resistance,  $R$ , in series with the source of the transconductance transistor, let the effective transconductance be

$$g_{m(eff)} = \frac{g_m}{1 + g_m R}$$



$$v_{gs2} = v_{g2} - v_{s2} = v_{in} - (g_{m2}r_{ds1})v_{gs2} \Rightarrow v_{gs2} = \frac{v_{in}}{1 + g_{m2}r_{ds1}}$$

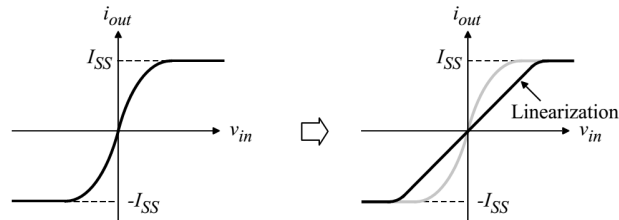
$$i_{out} = \frac{g_{m2}v_{in}}{1 + g_{m2}r_{ds1}} = g_{m2(eff)} v_{in}$$

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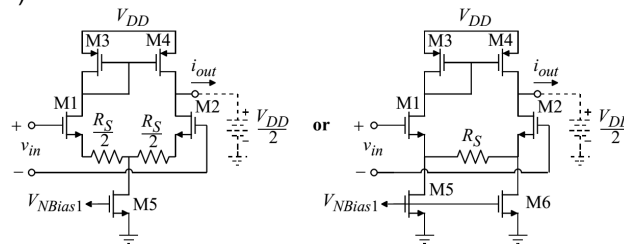
## LARGE SIGNAL PERFORMANCE OF THE DIFFERENTIAL AMPLIFIER

### Linearization of the Transconductance

- Goal:



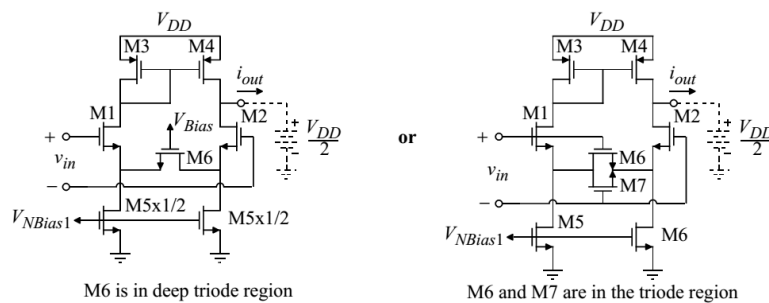
- Method (degeneration):



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## LARGE SIGNAL PERFORMANCE OF THE DIFFERENTIAL AMPLIFIER

### Linearization with Active Devices



M6 is in deep triode region

M6 and M7 are in the triode region

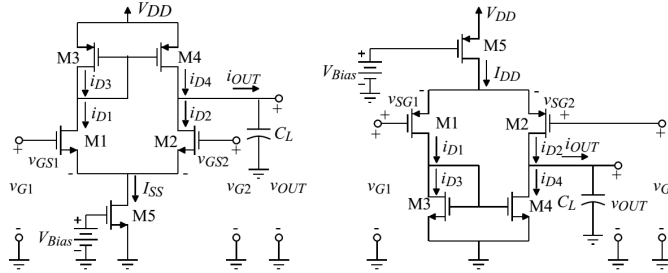
- Note that these transconductors on this slide and the last can all have a varying transconductance by changing the value of  $I_{SS}$ .

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## LARGE SIGNAL PERFORMANCE OF THE DIFFERENTIAL AMPLIFIER

### Slew Rate of the Differential Amplifier

- Slew Rate (SR) = Maximum output-voltage rate (either positive or negative)
- It is caused by,  $i_{OUT} = C_L \frac{dv_{OUT}}{dt}$ . When  $i_{OUT}$  is a constant, the rate is a constant.
- Consider the following current-mirror load, differential amplifiers:



- Note that slew rate can only occur when the differential input signal is large enough to cause  $I_{SS}$  ( $I_{DD}$ ) to flow through only one of the differential input transistors.

$$SR = \frac{I_{SS}}{C_L} = \frac{I_{DD}}{C_L} \rightarrow \text{If } C_L = 5\text{pF and } I_{SS} = 10\mu\text{A, the slew rate is } SR = 2\text{V}/\mu\text{s}$$

(For the BJT differential amplifier slewing occurs at  $\pm 100\text{mV}$  whereas for the MOSFET differential amplifier it can be  $\pm 2\text{V}$  or more.) 101

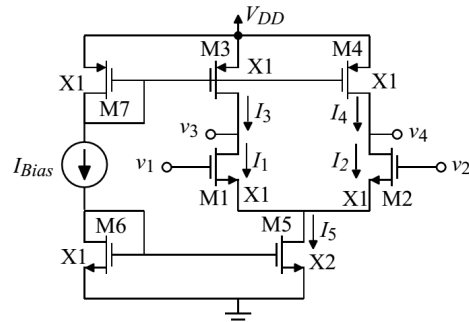
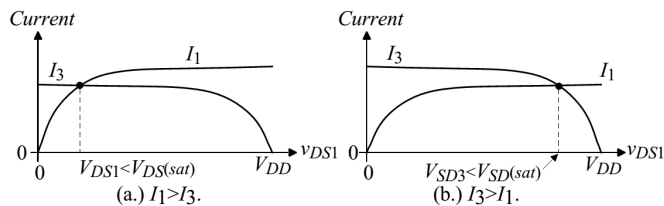
## DIFFERENTIAL AMPLIFIERS WITH CURRENT SOURCE LOADS

### Comments:

- Differential Amplifiers are not truly balanced  $\rightarrow$  Need to be improved
- The upper ICMR is low

### Current-Source Load Differential Amplifier

- Gives a truly balanced differential amplifier.
- Also, the upper input common-mode range is extended.
- A problem occurs if  $I_1 \neq I_3$  or if  $I_2 \neq I_4$  (current mismatch)



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## DESIGN OF DIFFERENTIAL AMPLIFIERS

### Design of a CMOS Differential Amplifier with a Current Mirror Load

#### Design Considerations

##### Constraints

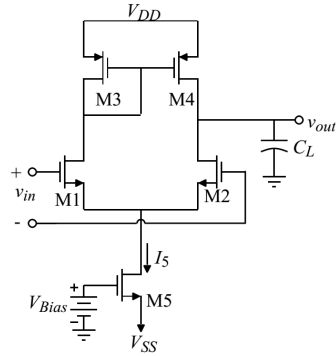
Power supply  
Technology  
Temperature

##### Specifications

Small-signal gain  
Frequency response  
Slew rate ( $C_L$ )  
Power dissipation

#### Relationships

- $A_v = g_{m1} R_{out}$
- $\omega_{-3dB} = 1/R_{out} C_L$
- $V_{IC(max)} = V_{DD} - V_{SG3} + V_{TN1}$
- $V_{IC(min)} = V_{SS} + V_{DS5(sat)} + V_{GS1} = V_{SS} + V_{DS5(sat)} + V_{GS2}$
- $SR = I_{SS}/C_L$
- $P_{diss} = (V_{DD} + |V_{SS}|) \times (\text{All dc currents flowing from } V_{DD} \text{ or to } V_{SS})$



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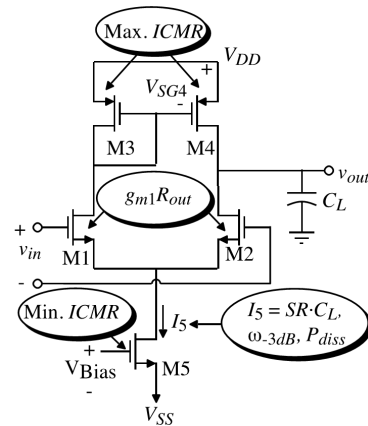
## DESIGN OF DIFFERENTIAL AMPLIFIERS

### Design of a CMOS Differential Amplifier with a Current Mirror Load

#### The design procedure is illustrated as (the right flow):

#### Procedure:

1. Pick  $I_{SS}$  to satisfy the slew rate knowing  $C_L$  or the power dissipation
2. Check to see if  $R_{out}$  will satisfy the frequency response, if not change  $I_{SS}$  or modify circuit
3. Design  $W_3/L_3$  ( $W_4/L_4$ ) to satisfy the upper  $ICMR$
4. Design  $W_1/L_1$  ( $W_2/L_2$ ) to satisfy the gain
5. Design  $W_5/L_5$  to satisfy the lower  $ICMR$
6. Iterate where necessary



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## DESIGN OF DIFFERENTIAL AMPLIFIERS

### Example - Design of a CMOS Differential Amplifier with a Current Mirror Load

- Design the currents and W/L values of the current mirror load MOS differential amplifier to satisfy the following specifications:  $V_{DD} = -V_{SS} = 2.5V$ ,  $SR \geq 10V/\mu s$  ( $C_L = 5pF$ ),  $f_{3dB} \geq 100kHz$  ( $C_L = 5pF$ ), a small signal gain of  $100V/V$ ,  $-1.5V \leq ICMR \leq 2V$  and  $P_{diss} \leq 1mW$ . Use the parameters of  $K_N' = 110\mu A/V^2$ ,  $K_P' = 50\mu A/V^2$ ,  $V_{TN} = 0.7V$ ,  $V_{TP} = -0.7V$ ,  $\lambda_N = 0.04V^{-1}$  and  $\lambda_P = 0.05V^{-1}$ .

#### Solution:

- To meet the slew rate,  $I_{SS} \geq 50\mu A$  ( $SL \cdot C_L$ ). For maximum  $P_{diss}$ ,  $I_{SS} \leq 200\mu A$  ( $\sim 1mW/(V_{DD} + |V_{SS}|)$ ).
- $f_{3dB}$  of 100kHz implies that  $R_{out} \leq 318k\Omega$ . Therefore,  $R_{out} = \frac{1}{(\lambda_N + \lambda_P)I_D} = \frac{2}{(\lambda_N + \lambda_P)I_{SS}} \leq 318k\Omega$   
 $I_{SS} \geq 70\mu A$ . Thus, pick  $I_{SS} = 100\mu A$
- $V_{IC(max)} = V_{DD} - V_{SG3} + V_{TN1} \rightarrow 2V = 2.5 - V_{SG3} + 0.7 \rightarrow V_{SG3} = 1.2V$

$$V_{SG3} = \sqrt{\frac{2 \cdot 50\mu A}{50\mu A/V^2 (W_3/L_3)}} + 0.7 = 1.2V \longrightarrow \frac{W_3}{L_3} = \frac{W_4}{L_4} = \frac{2}{(0.5)^2} = 8$$

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## DESIGN OF DIFFERENTIAL AMPLIFIERS

### Example - Design of a CMOS Differential Amplifier with a Current Mirror Load

$$4. \quad 100 = g_{m1}R_{out} = \frac{g_{m1}}{g_{ds2} + g_{ds4}} = \frac{\sqrt{2 \cdot 110\mu A/V^2 (W_1/L_1)}}{(0.04 + 0.05)\sqrt{50\mu A}} = 23.31 \sqrt{\frac{W_1}{L_1}} \rightarrow \frac{W_1}{L_1} = \frac{W_2}{L_2} = 18.4$$

$$5. \quad V_{IC(min)} = V_{SS} + V_{DS5(sat)} + V_{GS1}$$

$$-1.5 = -2.5 + V_{DS5(sat)} + \sqrt{\frac{2 \cdot 50\mu A}{50\mu A/V^2 (18.4)}} + 0.7$$

$$V_{DS5(sat)} = 0.3 - 0.222 = 0.078 \rightarrow \frac{W_5}{L_5} = \frac{2I_{SS}}{K_N' V_{DS5(sat)}^2} = 150.6$$

- We probably should increase  $W_1/L_1$  to reduce  $V_{GS1}$ . If we choose  $W_1/L_1 = 40$ , then  $V_{DS5(sat)} = 0.149V$  and  $W_5/L_5 = 41$ . (Larger than specified gain should be ok.)

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## DESIGN OF DIFFERENTIAL AMPLIFIERS

### SUMMARY

- Differential amplifiers are compatible with the matching properties of IC technology
- The differential amplifier has two modes of signal operation:
  - Differential mode
  - Common mode
- Differential amplifiers are excellent input stages for voltage amplifiers
- Differential amplifiers can have different loads including:
  - Current mirrors
  - MOS diodes
  - Current sources/sinks
  - Resistors
- The small signal performance of the differential amplifier is similar to the inverting amplifier in gain, output resistance and bandwidth
- The large signal performance includes slew rate and the linearization of the transconductance
- The design of CMOS analog circuits uses the relationships of the circuit to **design the DC currents and the W/L ratios of each transistor**.

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## DESIGN OF DIFFERENTIAL AMPLIFIERS

### SUMMARY

- W/L calculation

$$g_m = \sqrt{2K' \frac{W}{L} I_D}$$

$$i_D \approx \frac{1}{2} K' \left( \frac{W}{L} \right) v_{GS}^2$$

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**THE END  
(THIS PART)**

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